

# Data for the Pattern Recognition paper, highlighting the best formulas derived using the OMP regression model.

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## 1 Appendix A. Tables and formulas

Table 2 contains an example of the best formulas: (Tables 3, 4, 5, 6, 7, 8, 9, 10) were extracted from the OMP model and represent the 8 formulas with 49 terms. As can be seen in these two tables, Table 9 is the same as Table 1, demonstrating that our method successfully replicated the formula obtained inductively by Morgenthaler.

Additionally, we aim to demonstrate that this model yields more Euler equations beyond those with 49 terms. Specifically, we identified the following equations.

- Equations with 51 terms: accompanied by Tables 11, 12, 13, and 14.
- Equations with 52 terms: accompanied by Tables 15, 16, 17, and 18.
- Equations with 54 terms: accompanied by 19, 20, 21, and 22.
- Equations with 80 terms: accompanied by 23, 24, 25, and 26.

These equations yield the same Euler number as those with 49 terms, demonstrating the model's versatility and robustness.

Next, we present the formulas and tables for shape descriptors: contact surface, enclosing surface, and volume.

- Enclosing surface with 103 terms: accompanied by Tables 27 and 28.
- Enclosing surface with 103 terms: accompanied by Tables 29 and 30.
- Contact surface with 104 terms: accompanied by Tables 31 and 32.
- Contact surface with 104 terms: accompanied by Tables 33 and 34.
- Volume with 93 terms: accompanied by Tables 35 and 36.

- Volume with 93 terms: accompanied by Tables 37 and 38.

Finally, we demonstrate the feasibility of obtaining formulas using simplexes and tetravoxels.

- Points with 98 terms: accompanied by Tables 39 and 40.
- Points with 98 terms: accompanied by Tables 41 and 42.
- Edges with 113 terms: accompanied by Tables 43 and 44.
- Edges with 113 terms: accompanied by Tables 45 and 46.
- Faces with 114 terms: accompanied by Tables 47 and 48.
- Faces with 114 terms: accompanied by Tables 49 and 50.
- Tetra with 38 terms: accompanied by Tables 51, 52, 53, 54, 55, 56, 57, 58.

In addition, along with shape descriptors, simplexes, and tetra-voxels, the linear regression models demonstrated a remarkable accuracy of 100%. Nevertheless, considering the magnitude of the resulting formulae, the formulas derived from OMP produced the most favorable outcomes in terms of both metric and computational cost.

Table 1: Indexes of the Euler number increments of voxels patterns.

| <b>Index</b> | <b><math>\Delta E</math></b> | <b>Index</b> | <b><math>\Delta E</math></b> | <b>Index</b> | <b><math>\Delta E</math></b> | <b>Index</b> | <b><math>\Delta E</math></b> |
|--------------|------------------------------|--------------|------------------------------|--------------|------------------------------|--------------|------------------------------|
| 00000010     | 1.0                          | 00001001     | -1.0                         | 00001011     | -1.0                         | 00011000     | -1.0                         |
| 00011001     | -1.0                         | 00011010     | -1.0                         | 00011011     | -1.0                         | 00100001     | -1.0                         |
| 00100011     | -1.0                         | 00100100     | -1.0                         | 00100101     | -1.0                         | 00100110     | -1.0                         |
| 00100111     | -1.0                         | 00101000     | -1.0                         | 00101001     | -2.0                         | 00101010     | -1.0                         |
| 00101011     | -2.0                         | 00101100     | -1.0                         | 00101101     | -1.0                         | 00101110     | -1.0                         |
| 00101111     | -1.0                         | 00111000     | -1.0                         | 00111001     | -1.0                         | 00111010     | -1.0                         |
| 00111011     | -1.0                         | 10000001     | -1.0                         | 10000011     | -1.0                         | 10001001     | -1.0                         |
| 10001011     | -1.0                         | 10010100     | 1.0                          | 10010101     | 1.0                          | 10010110     | 1.0                          |
| 10010111     | 1.0                          | 10011100     | 1.0                          | 10011101     | 1.0                          | 10011110     | 1.0                          |
| 10011111     | 1.0                          | 10100001     | -1.0                         | 10100011     | -1.0                         | 10101001     | -1.0                         |
| 10101011     | -1.0                         | 10110100     | 1.0                          | 10110101     | 1.0                          | 10110110     | 1.0                          |
| 10110111     | 1.0                          | 10111100     | 1.0                          | 10111101     | 1.0                          | 10111110     | 1.0                          |
| 10111111     | 1.0                          | Others       | 0.0                          |              |                              |              |                              |

Table 2: The table compares the number of formulas derived for each descriptor using variable selection regression models. It identifies the most effective formulas with the fewest coefficients and specifies the model in which they were found.

| Model       | Descriptors and Simplexes | Statistics |     |        | $N^{\#}$<br>formulas found |
|-------------|---------------------------|------------|-----|--------|----------------------------|
|             |                           | Min        | Max | Median |                            |
| OMP         | Euler                     | <u>49</u>  | 134 | 85     | 111                        |
|             | ES                        | <u>103</u> | 179 | 136.5  | 1286                       |
|             | CS                        | <u>104</u> | 186 | 139    | 1303                       |
|             | Volume                    | <u>93</u>  | 178 | 126    | 1010                       |
|             | Edges                     | <u>113</u> | 204 | 154.5  | 1414                       |
|             | Edges                     | <u>98</u>  | 202 | 145    | 1331                       |
|             | Faces                     | <u>114</u> | 194 | 150    | 1372                       |
|             | Tetravoxels               | <u>38</u>  | 38  | 38     | 8                          |
| ARD         | Euler                     | 80         | 208 | 167.5  | 474                        |
|             | ES                        | 111        | 254 | 201    | 29                         |
|             | CS                        | 112        | 221 | 160    | 154                        |
|             | Volume                    | 101        | 255 | 206    | 254                        |
|             | Points                    | 140        | 251 | 212    | 64                         |
|             | Edges                     | 163        | 255 | 199    | 28                         |
|             | Faces                     | 154        | 255 | 202    | 35                         |
|             | Tetravoxels               | <u>38</u>  | 63  | 52     | 12                         |
| LassoLars   | Euler                     | 80         | 201 | 138    | 320                        |
|             | ES                        | 105        | 217 | 161.5  | 3065                       |
|             | CS                        | 105        | 203 | 154    | 3134                       |
|             | Volume                    | 94         | 218 | 158    | 3022                       |
|             | Points                    | 115        | 218 | 167    | 3059                       |
|             | Edges                     | 124        | 218 | 173    | 2597                       |
|             | Faces                     | 119        | 218 | 169.5  | 2934                       |
|             | Tetravoxels               | <u>38</u>  | 86  | 54     | 466                        |
| LassoLarsIC | Euler                     | 80         | 196 | 149.5  | 280                        |
|             | ES                        | 105        | 216 | 161    | 3065                       |
|             | CS                        | 105        | 202 | 153.5  | 3142                       |
|             | Volume                    | 94         | 218 | 157.5  | 3030                       |
|             | Points                    | 124        | 220 | 174    | 3065                       |
|             | Edges                     | 124        | 218 | 173    | 2599                       |
|             | Faces                     | 119        | 218 | 169    | 2929                       |
|             | Tetravoxels               | <u>38</u>  | 63  | 51     | 273                        |

## 1.1 Tables from Euler using OMP Regression model with 49 coefficients

Table 3: Euler, Length: 49, Numbers: Number 0.0: 206, Number -1.0: 30, Number 1.0: 17, Number -2.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00010000 | 1.0        | 00100001 | -1.0       | 00100100 | -1.0       | 00100101 | -1.0       |
| 00110001 | -1.0       | 00110100 | -1.0       | 00110101 | -1.0       | 01000001 | -1.0       |
| 01000010 | -1.0       | 01000011 | -1.0       | 01010001 | -1.0       | 01010010 | -1.0       |
| 01010011 | -1.0       | 01100000 | -1.0       | 01100001 | -2.0       | 01100010 | -1.0       |
| 01100011 | -1.0       | 01100100 | -1.0       | 01100101 | -1.0       | 01110000 | -1.0       |
| 01110001 | -2.0       | 01110010 | -1.0       | 01110011 | -1.0       | 01110100 | -1.0       |
| 01110101 | -1.0       | 10000001 | -1.0       | 10000110 | 1.0        | 10000111 | 1.0        |
| 10010001 | -1.0       | 10010110 | 1.0        | 10010111 | 1.0        | 10100001 | -1.0       |
| 10100110 | 1.0        | 10100111 | 1.0        | 10110001 | -1.0       | 10110110 | 1.0        |
| 10110111 | 1.0        | 11000001 | -1.0       | 11000110 | 1.0        | 11000111 | 1.0        |
| 11010001 | -1.0       | 11010110 | 1.0        | 11010111 | 1.0        | 11100001 | -1.0       |
| 11100110 | 1.0        | 11100111 | 1.0        | 11110001 | -1.0       | 11110110 | 1.0        |
| 11110111 | 1.0        | Others   | 0.0        |          |            |          |            |

Table 4: Euler, Length: 49, Numbers: Number 0.0: 206, Number -1.0: 30, Number 1.0: 17, Number -2.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00010100 | -1.0       | 00011000 | -1.0       | 00011100 | -1.0       | 00100100 | -1.0       |
| 00101001 | 1.0        | 00101101 | 1.0        | 00110100 | -1.0       | 00111001 | 1.0        |
| 00111101 | 1.0        | 01000000 | 1.0        | 01010100 | -1.0       | 01011000 | -1.0       |
| 01011100 | -1.0       | 01100100 | -1.0       | 01101001 | 1.0        | 01101101 | 1.0        |
| 01110100 | -1.0       | 01111001 | 1.0        | 01111101 | 1.0        | 10000001 | -1.0       |
| 10000100 | -1.0       | 10000101 | -1.0       | 10010000 | -1.0       | 10010001 | -1.0       |
| 10010100 | -2.0       | 10010101 | -1.0       | 10011000 | -1.0       | 10011100 | -1.0       |
| 10100100 | -1.0       | 10101001 | 1.0        | 10101101 | 1.0        | 10110100 | -1.0       |
| 10111001 | 1.0        | 10111101 | 1.0        | 11000001 | -1.0       | 11000100 | -1.0       |
| 11000101 | -1.0       | 11010000 | -1.0       | 11010001 | -1.0       | 11010100 | -2.0       |
| 11010101 | -1.0       | 11011000 | -1.0       | 11011100 | -1.0       | 11100100 | -1.0       |
| 11101001 | 1.0        | 11101101 | 1.0        | 11110100 | -1.0       | 11111001 | 1.0        |
| 11111101 | 1.0        | Others   | 0.0        |          |            |          |            |

Table 5: Euler, Length: 49, Numbers: Number 0.0: 206, Number -1.0: 30, Number 1.0: 17, Number -2.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00010110 | 1.0        | 00011000 | -1.0       | 00011110 | 1.0        | 00100100 | -1.0       |
| 00101000 | -1.0       | 00101100 | -1.0       | 00110110 | 1.0        | 00111000 | -1.0       |
| 00111110 | 1.0        | 01000010 | -1.0       | 01001000 | -1.0       | 01001010 | -1.0       |
| 01010110 | 1.0        | 01011000 | -1.0       | 01011110 | 1.0        | 01100000 | -1.0       |
| 01100010 | -1.0       | 01100100 | -1.0       | 01101000 | -2.0       | 01101010 | -1.0       |
| 01101100 | -1.0       | 01110110 | 1.0        | 01111000 | -1.0       | 01111110 | 1.0        |
| 10000000 | 1.0        | 10010110 | 1.0        | 10011000 | -1.0       | 10011110 | 1.0        |
| 10100100 | -1.0       | 10101000 | -1.0       | 10101100 | -1.0       | 10110110 | 1.0        |
| 10111000 | -1.0       | 10111110 | 1.0        | 11000010 | -1.0       | 11001000 | -1.0       |
| 11001010 | -1.0       | 11010110 | 1.0        | 11011000 | -1.0       | 11011110 | 1.0        |
| 11100000 | -1.0       | 11100010 | -1.0       | 11100100 | -1.0       | 11101000 | -2.0       |
| 11101010 | -1.0       | 11101100 | -1.0       | 11110110 | 1.0        | 11111000 | -1.0       |
| 11111110 | 1.0        | Others   | 0.0        |          |            |          |            |

Table 6: Euler, Length: 49, Numbers: Number 0.0: 206, Number -1.0: 30, Number 1.0: 17, Number -2.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000110 | -1.0       | 00001000 | 1.0        | 00001110 | -1.0       | 00100100 | -1.0       |
| 00100110 | -1.0       | 00101100 | -1.0       | 00101110 | -1.0       | 01000010 | -1.0       |
| 01000110 | -1.0       | 01001010 | -1.0       | 01001110 | -1.0       | 01100001 | 1.0        |
| 01100011 | 1.0        | 01100101 | 1.0        | 01100111 | 1.0        | 01101001 | 1.0        |
| 01101011 | 1.0        | 01101101 | 1.0        | 01101111 | 1.0        | 10000001 | -1.0       |
| 10000010 | -1.0       | 10000011 | -1.0       | 10000100 | -1.0       | 10000101 | -1.0       |
| 10000110 | -2.0       | 10000111 | -1.0       | 10001001 | -1.0       | 10001010 | -1.0       |
| 10001011 | -1.0       | 10001100 | -1.0       | 10001101 | -1.0       | 10001110 | -2.0       |
| 10001111 | -1.0       | 10100100 | -1.0       | 10100110 | -1.0       | 10101100 | -1.0       |
| 10101110 | -1.0       | 11000010 | -1.0       | 11000110 | -1.0       | 11001010 | -1.0       |
| 11001110 | -1.0       | 11100001 | 1.0        | 11100011 | 1.0        | 11100101 | 1.0        |
| 11100111 | 1.0        | 11101001 | 1.0        | 11101011 | 1.0        | 11101101 | 1.0        |
| 11101111 | 1.0        | Others   | 0.0        |          |            |          |            |

Table 7: Euler, Length: 49, Numbers: Number 0.0: 206, Number -1.0: 30, Number 1.0: 17, Number -2.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00010010 | -1.0       | 00011000 | -1.0       | 00011010 | -1.0       | 00100000 | 1.0        |
| 00110010 | -1.0       | 00111000 | -1.0       | 00111010 | -1.0       | 01000010 | -1.0       |
| 01001001 | 1.0        | 01001011 | 1.0        | 01010010 | -1.0       | 01011001 | 1.0        |
| 01011011 | 1.0        | 01100010 | -1.0       | 01101001 | 1.0        | 01101011 | 1.0        |
| 01110010 | -1.0       | 01111001 | 1.0        | 01111011 | 1.0        | 10000001 | -1.0       |
| 10000010 | -1.0       | 10000011 | -1.0       | 10010000 | -1.0       | 10010001 | -1.0       |
| 10010010 | -2.0       | 10010011 | -1.0       | 10011000 | -1.0       | 10011010 | -1.0       |
| 10100001 | -1.0       | 10100010 | -1.0       | 10100011 | -1.0       | 10110000 | -1.0       |
| 10110001 | -1.0       | 10110010 | -2.0       | 10110011 | -1.0       | 10111000 | -1.0       |
| 10111010 | -1.0       | 11000010 | -1.0       | 11001001 | 1.0        | 11001011 | 1.0        |
| 11010010 | -1.0       | 11011001 | 1.0        | 11011011 | 1.0        | 11100010 | -1.0       |
| 11101001 | 1.0        | 11101011 | 1.0        | 11110010 | -1.0       | 11111001 | 1.0        |
| 11111011 | 1.0        | Others   | 0.0        |          |            |          |            |

Table 8: Euler, Length: 49, Numbers: Number 0.0: 206, Number -1.0: 30, Number 1.0: 17, Number -2.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000100 | 1.0        | 00001001 | -1.0       | 00001101 | -1.0       | 00011000 | -1.0       |
| 00011001 | -1.0       | 00011100 | -1.0       | 00011101 | -1.0       | 01000001 | -1.0       |
| 01000010 | -1.0       | 01000011 | -1.0       | 01000101 | -1.0       | 01000110 | -1.0       |
| 01000111 | -1.0       | 01001000 | -1.0       | 01001001 | -2.0       | 01001010 | -1.0       |
| 01001011 | -1.0       | 01001100 | -1.0       | 01001101 | -2.0       | 01001110 | -1.0       |
| 01001111 | -1.0       | 01011000 | -1.0       | 01011001 | -1.0       | 01011100 | -1.0       |
| 01011101 | -1.0       | 10000001 | -1.0       | 10000101 | -1.0       | 10001001 | -1.0       |
| 10001101 | -1.0       | 10010010 | 1.0        | 10010011 | 1.0        | 10010110 | 1.0        |
| 10010111 | 1.0        | 10011010 | 1.0        | 10011011 | 1.0        | 10011110 | 1.0        |
| 10011111 | 1.0        | 11000001 | -1.0       | 11000101 | -1.0       | 11001001 | -1.0       |
| 11001101 | -1.0       | 11010010 | 1.0        | 11010011 | 1.0        | 11010110 | 1.0        |
| 11010111 | 1.0        | 11011010 | 1.0        | 11011011 | 1.0        | 11011110 | 1.0        |
| 11011111 | 1.0        | Others   | 0.0        |          |            |          |            |

Table 9: Euler, Length: 49, Numbers: Number 0.0: 206, Number -1.0: 30, Number 1.0: 17, Number -2.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 1.0        | 00001001 | -1.0       | 00001011 | -1.0       | 00011000 | -1.0       |
| 00011001 | -1.0       | 00011010 | -1.0       | 00011011 | -1.0       | 00100001 | -1.0       |
| 00100011 | -1.0       | 00100100 | -1.0       | 00100101 | -1.0       | 00100110 | -1.0       |
| 00100111 | -1.0       | 00101000 | -1.0       | 00101001 | -2.0       | 00101010 | -1.0       |
| 00101011 | -2.0       | 00101100 | -1.0       | 00101101 | -1.0       | 00101110 | -1.0       |
| 00101111 | -1.0       | 00111000 | -1.0       | 00111001 | -1.0       | 00111010 | -1.0       |
| 00111011 | -1.0       | 10000001 | -1.0       | 10000011 | -1.0       | 10001001 | -1.0       |
| 10001011 | -1.0       | 10010100 | 1.0        | 10010101 | 1.0        | 10010110 | 1.0        |
| 10010111 | 1.0        | 10011100 | 1.0        | 10011101 | 1.0        | 10011110 | 1.0        |
| 10011111 | 1.0        | 10100001 | -1.0       | 10100011 | -1.0       | 10101001 | -1.0       |
| 10101011 | -1.0       | 10110100 | 1.0        | 10110101 | 1.0        | 10110110 | 1.0        |
| 10110111 | 1.0        | 10111100 | 1.0        | 10111101 | 1.0        | 10111110 | 1.0        |
| 10111111 | 1.0        | Others   | 0.0        |          |            |          |            |

Table 10: Euler, Length: 49, Numbers: Number 0.0: 206, Number -1.0: 30, Number 1.0: 17, Number -2.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | 1.0        | 00000110 | -1.0       | 00000111 | -1.0       | 00010010 | -1.0       |
| 00010011 | -1.0       | 00010100 | -1.0       | 00010101 | -1.0       | 00010110 | -2.0       |
| 00010111 | -2.0       | 00011000 | -1.0       | 00011001 | -1.0       | 00011010 | -1.0       |
| 00011011 | -1.0       | 00011100 | -1.0       | 00011101 | -1.0       | 00011110 | -1.0       |
| 00011111 | -1.0       | 00100100 | -1.0       | 00100101 | -1.0       | 00100110 | -1.0       |
| 00100111 | -1.0       | 00110100 | -1.0       | 00110101 | -1.0       | 00110110 | -1.0       |
| 00110111 | -1.0       | 01000010 | -1.0       | 01000011 | -1.0       | 01000110 | -1.0       |
| 01000111 | -1.0       | 01010010 | -1.0       | 01010011 | -1.0       | 01010110 | -1.0       |
| 01010111 | -1.0       | 01101000 | 1.0        | 01101001 | 1.0        | 01101010 | 1.0        |
| 01101011 | 1.0        | 01101100 | 1.0        | 01101101 | 1.0        | 01101110 | 1.0        |
| 01110111 | 1.0        | 01111000 | 1.0        | 01111001 | 1.0        | 01111010 | 1.0        |
| 01111011 | 1.0        | 01111100 | 1.0        | 01111101 | 1.0        | 01111110 | 1.0        |
| 01111111 | 1.0        | Others   | 0.0        |          |            |          |            |

## 1.2 Tables from Euler using OMP Regression model with 51 coefficients

Table 11: Euler, Length: 51, Numbers: Number 0.0: 204, Number -1.0: 34, Number 1.0: 17

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00010010 | -1.0       | 00011000 | -1.0       | 00011010 | -1.0       | 00100000 | 1.0        |
| 00110010 | -1.0       | 00111000 | -1.0       | 00111010 | -1.0       | 01000001 | -1.0       |
| 01000010 | -1.0       | 01000011 | -1.0       | 01010001 | -1.0       | 01010010 | -1.0       |
| 01010011 | -1.0       | 01100001 | -1.0       | 01100010 | -1.0       | 01100011 | -1.0       |
| 01110001 | -1.0       | 01110010 | -1.0       | 01110011 | -1.0       | 10000001 | -1.0       |
| 10000110 | 1.0        | 10000111 | 1.0        | 10010000 | -1.0       | 10010001 | -1.0       |
| 10010010 | -1.0       | 10010110 | 1.0        | 10010111 | 1.0        | 10011000 | -1.0       |
| 10011010 | -1.0       | 10100001 | -1.0       | 10100110 | 1.0        | 10100111 | 1.0        |
| 10110000 | -1.0       | 10110001 | -1.0       | 10110010 | -1.0       | 10110110 | 1.0        |
| 10110111 | 1.0        | 10111000 | -1.0       | 10111010 | -1.0       | 11000001 | -1.0       |
| 11000110 | 1.0        | 11000111 | 1.0        | 11010001 | -1.0       | 11010110 | 1.0        |
| 11010111 | 1.0        | 11100001 | -1.0       | 11100110 | 1.0        | 11100111 | 1.0        |
| 11110001 | -1.0       | 11110110 | 1.0        | 11110111 | 1.0        | Others   | 0.0        |

Table 12: Euler, Length: 51, Numbers: Number 0.0: 204, Number -1.0: 34, Number 1.0: 17

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | 1.0        | 00000110 | -1.0       | 00000111 | -1.0       | 00010100 | -1.0       |
| 00010101 | -1.0       | 00010110 | -1.0       | 00010111 | -1.0       | 00011000 | -1.0       |
| 00011001 | -1.0       | 00011100 | -1.0       | 00011101 | -1.0       | 00100100 | -1.0       |
| 00100101 | -1.0       | 00100110 | -1.0       | 00100111 | -1.0       | 00110100 | -1.0       |
| 00110101 | -1.0       | 00110110 | -1.0       | 00110111 | -1.0       | 01000010 | -1.0       |
| 01000011 | -1.0       | 01000110 | -1.0       | 01000111 | -1.0       | 01001000 | -1.0       |
| 01001001 | -1.0       | 01001010 | -1.0       | 01001011 | -1.0       | 01001100 | -1.0       |
| 01001101 | -1.0       | 01001110 | -1.0       | 01001111 | -1.0       | 01011000 | -1.0       |
| 01011001 | -1.0       | 01011100 | -1.0       | 01011101 | -1.0       | 10010010 | 1.0        |
| 10010011 | 1.0        | 10010110 | 1.0        | 10010111 | 1.0        | 10011010 | 1.0        |
| 10011011 | 1.0        | 10011110 | 1.0        | 10011111 | 1.0        | 11010010 | 1.0        |
| 11010011 | 1.0        | 11010110 | 1.0        | 11010111 | 1.0        | 11011010 | 1.0        |
| 11011011 | 1.0        | 11011110 | 1.0        | 11011111 | 1.0        | Others   | 0.0        |



Table 13: Euler, Length: 51, Numbers: Number 0.0: 204, Number -1.0: 34, Number 1.0: 17

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000110 | -1.0       | 00001000 | 1.0        | 00001110 | -1.0       | 00100100 | -1.0       |
| 00100110 | -1.0       | 00101100 | -1.0       | 00101110 | -1.0       | 01000001 | -1.0       |
| 01000010 | -1.0       | 01000011 | -1.0       | 01000101 | -1.0       | 01000110 | -1.0       |
| 01000111 | -1.0       | 01001001 | -1.0       | 01001010 | -1.0       | 01001011 | -1.0       |
| 01001101 | -1.0       | 01001110 | -1.0       | 01001111 | -1.0       | 10000001 | -1.0       |
| 10000100 | -1.0       | 10000101 | -1.0       | 10000110 | -1.0       | 10001001 | -1.0       |
| 10001100 | -1.0       | 10001101 | -1.0       | 10001110 | -1.0       | 10010010 | 1.0        |
| 10010011 | 1.0        | 10010110 | 1.0        | 10010111 | 1.0        | 10011010 | 1.0        |
| 10011011 | 1.0        | 10011110 | 1.0        | 10011111 | 1.0        | 10100100 | -1.0       |
| 10100110 | -1.0       | 10101100 | -1.0       | 10101110 | -1.0       | 11000001 | -1.0       |
| 11000101 | -1.0       | 11001001 | -1.0       | 11001101 | -1.0       | 11010010 | 1.0        |
| 11010011 | 1.0        | 11010110 | 1.0        | 11010111 | 1.0        | 11011010 | 1.0        |
| 11011011 | 1.0        | 11011110 | 1.0        | 11011111 | 1.0        | Others   | 0.0        |

Table 14: Euler, Length: 51, Numbers: Number 0.0: 204, Number -1.0: 34, Number 1.0: 17

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00010110 | 1.0        | 00011000 | -1.0       | 00011110 | 1.0        | 00100100 | -1.0       |
| 00101000 | -1.0       | 00101100 | -1.0       | 00110110 | 1.0        | 00111000 | -1.0       |
| 00111110 | 1.0        | 01000000 | 1.0        | 01010110 | 1.0        | 01011000 | -1.0       |
| 01011110 | 1.0        | 01100100 | -1.0       | 01101000 | -1.0       | 01101100 | -1.0       |
| 01110110 | 1.0        | 01111000 | -1.0       | 01111110 | 1.0        | 10000001 | -1.0       |
| 10000100 | -1.0       | 10000101 | -1.0       | 10010000 | -1.0       | 10010001 | -1.0       |
| 10010100 | -1.0       | 10010101 | -1.0       | 10010110 | 1.0        | 10011000 | -1.0       |
| 10011110 | 1.0        | 10100100 | -1.0       | 10101000 | -1.0       | 10101100 | -1.0       |
| 10110110 | 1.0        | 10111000 | -1.0       | 10111110 | 1.0        | 11000001 | -1.0       |
| 11000100 | -1.0       | 11000101 | -1.0       | 11010000 | -1.0       | 11010001 | -1.0       |
| 11010100 | -1.0       | 11010101 | -1.0       | 11010110 | 1.0        | 11011000 | -1.0       |
| 11011110 | 1.0        | 11100100 | -1.0       | 11101000 | -1.0       | 11101100 | -1.0       |
| 11110110 | 1.0        | 11111000 | -1.0       | 11111110 | 1.0        | Others   | 0.0        |

### 1.3 Tables from Euler using OMP Regression model with 52 coefficients

Table 15: Euler, Length: 52, Numbers: Number 0.0: 203, Number -1.0: 33, Number 1.0: 18, Number -2.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000100 | 1.0        | 00001001 | -1.0       | 00001101 | -1.0       | 00011000 | -1.0       |
| 00011001 | -1.0       | 00011100 | -1.0       | 00011101 | -1.0       | 01000001 | -1.0       |
| 01000010 | -1.0       | 01000011 | -1.0       | 01000110 | -1.0       | 01001000 | -1.0       |
| 01001001 | -2.0       | 01001010 | -1.0       | 01001011 | -1.0       | 01001100 | -1.0       |
| 01001101 | -1.0       | 01001110 | -1.0       | 01011000 | -1.0       | 01011001 | -1.0       |
| 01011100 | -1.0       | 01011101 | -1.0       | 01100101 | 1.0        | 01100111 | 1.0        |
| 01101101 | 1.0        | 01101111 | 1.0        | 10000001 | -1.0       | 10000101 | -1.0       |
| 10001001 | -1.0       | 10001010 | -1.0       | 10001011 | -1.0       | 10001101 | -1.0       |
| 10001110 | -1.0       | 10001111 | -1.0       | 10010010 | 1.0        | 10010011 | 1.0        |
| 10010110 | 1.0        | 10010111 | 1.0        | 11000001 | -1.0       | 11000111 | 1.0        |
| 11001001 | -1.0       | 11001010 | -1.0       | 11001011 | -1.0       | 11001110 | -1.0       |
| 11010010 | 1.0        | 11010011 | 1.0        | 11010110 | 1.0        | 11010111 | 1.0        |
| 11100101 | 1.0        | 11100111 | 1.0        | 11101101 | 1.0        | 11101111 | 1.0        |
| Others   | 0.0        |          |            |          |            |          |            |

Table 16: Euler, Length: 52, Numbers: Number 0.0: 203, Number -1.0: 33, Number 1.0: 18, Number -2.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | 1.0        | 00000110 | -1.0       | 00000111 | -1.0       | 00010010 | -1.0       |
| 00010100 | -1.0       | 00010101 | -1.0       | 00010110 | -2.0       | 00010111 | -1.0       |
| 00011000 | -1.0       | 00011001 | -1.0       | 00011010 | -1.0       | 00011100 | -1.0       |
| 00011101 | -1.0       | 00011110 | -1.0       | 00100100 | -1.0       | 00100101 | -1.0       |
| 00100110 | -1.0       | 00100111 | -1.0       | 00110100 | -1.0       | 00110101 | -1.0       |
| 00110110 | -1.0       | 00110111 | -1.0       | 01000010 | -1.0       | 01000011 | -1.0       |
| 01000110 | -1.0       | 01000111 | -1.0       | 01001100 | -1.0       | 01001101 | -1.0       |
| 01001110 | -1.0       | 01001111 | -1.0       | 01010010 | -1.0       | 01010110 | -1.0       |
| 01011011 | 1.0        | 01011100 | -1.0       | 01011101 | -1.0       | 01011110 | -1.0       |
| 01101000 | 1.0        | 01101001 | 1.0        | 01101010 | 1.0        | 01101011 | 1.0        |
| 01111000 | 1.0        | 01111001 | 1.0        | 01111010 | 1.0        | 01111011 | 1.0        |
| 10010011 | 1.0        | 10010111 | 1.0        | 10011011 | 1.0        | 10011111 | 1.0        |
| 11010011 | 1.0        | 11010111 | 1.0        | 11011011 | 1.0        | 11011111 | 1.0        |
| Others   | 0.0        |          |            |          |            |          |            |

Table 17: Euler, Length: 52, Numbers: Number 0.0: 203, Number -1.0: 33, Number 1.0: 18, Number -2.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000100 | 1.0        | 00001101 | -1.0       | 00011000 | -1.0       | 00011100 | -1.0       |
| 00011101 | -1.0       | 00101001 | 1.0        | 00111001 | 1.0        | 01000001 | -1.0       |
| 01000010 | -1.0       | 01000011 | -1.0       | 01000101 | -1.0       | 01000110 | -1.0       |
| 01000111 | -1.0       | 01001000 | -1.0       | 01001001 | -1.0       | 01001010 | -1.0       |
| 01001011 | -1.0       | 01001100 | -1.0       | 01001101 | -2.0       | 01001110 | -1.0       |
| 01001111 | -1.0       | 01011000 | -1.0       | 01011100 | -1.0       | 01011101 | -1.0       |
| 01101001 | 1.0        | 01111001 | 1.0        | 10000001 | -1.0       | 10000101 | -1.0       |
| 10001101 | -1.0       | 10010000 | -1.0       | 10010001 | -1.0       | 10010100 | -1.0       |
| 10010101 | -1.0       | 10011000 | -1.0       | 10011100 | -1.0       | 10011101 | -1.0       |
| 10101001 | 1.0        | 10111001 | 1.0        | 11000001 | -1.0       | 11000101 | -1.0       |
| 11001101 | -1.0       | 11010010 | 1.0        | 11010011 | 1.0        | 11010110 | 1.0        |
| 11010111 | 1.0        | 11011001 | 1.0        | 11011010 | 1.0        | 11011011 | 1.0        |
| 11011110 | 1.0        | 11011111 | 1.0        | 11101001 | 1.0        | 11111001 | 1.0        |
| Others   | 0.0        |          |            |          |            |          |            |

Table 18: Euler, Length: 52, Numbers: Number 0.0: 203, Number -1.0: 33, Number 1.0: 18, Number -2.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000111 | -1.0       | 00010000 | 1.0        | 00010111 | -1.0       | 00100001 | -1.0       |
| 00100100 | -1.0       | 00100101 | -1.0       | 00100111 | -1.0       | 00110001 | -1.0       |
| 00110100 | -1.0       | 00110101 | -1.0       | 00110111 | -1.0       | 01000001 | -1.0       |
| 01000010 | -1.0       | 01000011 | -1.0       | 01000111 | -1.0       | 01010001 | -1.0       |
| 01010010 | -1.0       | 01010011 | -1.0       | 01010111 | -1.0       | 01100000 | -1.0       |
| 01100001 | -2.0       | 01100010 | -1.0       | 01100011 | -1.0       | 01100100 | -1.0       |
| 01100101 | -1.0       | 01100111 | -1.0       | 01110001 | -1.0       | 01110110 | 1.0        |
| 01111000 | 1.0        | 01111001 | 1.0        | 01111010 | 1.0        | 01111011 | 1.0        |
| 01111100 | 1.0        | 01111101 | 1.0        | 01111110 | 1.0        | 01111111 | 1.0        |
| 10000001 | -1.0       | 10000110 | 1.0        | 10010001 | -1.0       | 10010110 | 1.0        |
| 10100001 | -1.0       | 10100110 | 1.0        | 10110001 | -1.0       | 10110110 | 1.0        |
| 11000001 | -1.0       | 11000110 | 1.0        | 11010001 | -1.0       | 11010110 | 1.0        |
| 11100001 | -1.0       | 11100110 | 1.0        | 11110001 | -1.0       | 11110110 | 1.0        |
| Others   | 0.0        |          |            |          |            |          |            |

## 1.4 Tables from Euler using OMP Regression model with 54 coefficients

Table 19: Euler, Length: 54, Numbers: Number 0.0: 201, Number -1.0: 31, Number 1.0: 20, Number -2.0: 3

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000100 | 1.0        | 00001000 | -1.0       | 00001001 | -1.0       | 00001101 | -1.0       |
| 00011000 | -2.0       | 00011001 | -1.0       | 00011100 | -1.0       | 00011101 | -1.0       |
| 00101000 | -1.0       | 00111000 | -1.0       | 01000010 | -1.0       | 01000110 | -1.0       |
| 01001000 | -2.0       | 01001001 | -1.0       | 01001010 | -1.0       | 01001100 | -1.0       |
| 01001101 | -1.0       | 01001110 | -1.0       | 01011000 | -2.0       | 01011001 | -1.0       |
| 01011100 | -1.0       | 01011101 | -1.0       | 01100001 | 1.0        | 01100011 | 1.0        |
| 01100101 | 1.0        | 01100111 | 1.0        | 01101000 | -1.0       | 01101001 | 1.0        |
| 01101011 | 1.0        | 01101101 | 1.0        | 01101111 | 1.0        | 01111000 | -1.0       |
| 10000000 | 1.0        | 10000100 | 1.0        | 10001100 | 1.0        | 10011000 | -1.0       |
| 10101000 | -1.0       | 10111000 | -1.0       | 11000010 | -1.0       | 11000110 | -1.0       |
| 11001000 | -1.0       | 11001010 | -1.0       | 11001110 | -1.0       | 11011000 | -1.0       |
| 11100001 | 1.0        | 11100011 | 1.0        | 11100101 | 1.0        | 11100111 | 1.0        |
| 11101000 | -1.0       | 11101001 | 1.0        | 11101011 | 1.0        | 11101101 | 1.0        |
| 11101111 | 1.0        | 11111000 | -1.0       | Others   | 0.0        |          |            |

Table 20: Euler, Length: 54, Numbers: Number 0.0: 201, Number -1.0: 31, Number 1.0: 20, Number -2.0: 3

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 1.0        | 00010000 | 1.0        | 00010010 | 1.0        | 00100000 | -1.0       |
| 00100001 | -2.0       | 00100011 | -1.0       | 00100100 | -2.0       | 00100101 | -2.0       |
| 00100110 | -1.0       | 00100111 | -1.0       | 00101000 | -1.0       | 00101001 | -1.0       |
| 00101010 | -1.0       | 00101011 | -1.0       | 00101100 | -1.0       | 00101101 | -1.0       |
| 00101110 | -1.0       | 00101111 | -1.0       | 00110001 | -1.0       | 00110010 | 1.0        |
| 00110100 | -1.0       | 00110101 | -1.0       | 01001001 | 1.0        | 01001011 | 1.0        |
| 01011001 | 1.0        | 01011011 | 1.0        | 01100000 | -1.0       | 01100001 | -1.0       |
| 01100100 | -1.0       | 01100101 | -1.0       | 01101001 | 1.0        | 01101011 | 1.0        |
| 01110000 | -1.0       | 01110001 | -1.0       | 01110100 | -1.0       | 01110101 | -1.0       |
| 01111001 | 1.0        | 01111011 | 1.0        | 10000001 | -1.0       | 10000011 | -1.0       |
| 10010001 | -1.0       | 10010011 | -1.0       | 10100001 | -1.0       | 10100011 | -1.0       |
| 10110001 | -1.0       | 10110011 | -1.0       | 11001001 | 1.0        | 11001011 | 1.0        |
| 11011001 | 1.0        | 11011011 | 1.0        | 11101001 | 1.0        | 11101011 | 1.0        |
| 11111001 | 1.0        | 11111011 | 1.0        | Others   | 0.0        |          |            |

Table 21: Euler, Length: 54, Numbers: Number 0.0: 201, Number -1.0: 31, Number 1.0: 20, Number -2.0: 3

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | -1.0       | 00000010 | 1.0        | 00000100 | 1.0        | 00000110 | 1.0        |
| 00000111 | 1.0        | 00001001 | -2.0       | 00001011 | -1.0       | 00001101 | -1.0       |
| 00011000 | -1.0       | 00011001 | -1.0       | 00011010 | -1.0       | 00011011 | -1.0       |
| 00011100 | -1.0       | 00011101 | -1.0       | 00011110 | -1.0       | 00011111 | -1.0       |
| 00100001 | -1.0       | 00100011 | -1.0       | 00101001 | -1.0       | 00101011 | -1.0       |
| 01000001 | -1.0       | 01000101 | -1.0       | 01001001 | -1.0       | 01001101 | -1.0       |
| 01101000 | 1.0        | 01101001 | 1.0        | 01101010 | 1.0        | 01101011 | 1.0        |
| 01101100 | 1.0        | 01101101 | 1.0        | 01101110 | 1.0        | 01101111 | 1.0        |
| 01111000 | 1.0        | 01111001 | 1.0        | 01111010 | 1.0        | 01111011 | 1.0        |
| 01111100 | 1.0        | 01111101 | 1.0        | 01111110 | 1.0        | 01111111 | 1.0        |
| 10000001 | -2.0       | 10000011 | -1.0       | 10000101 | -1.0       | 10001001 | -2.0       |
| 10001011 | -1.0       | 10001101 | -1.0       | 10100001 | -1.0       | 10100011 | -1.0       |
| 10101001 | -1.0       | 10101011 | -1.0       | 11000001 | -1.0       | 11000101 | -1.0       |
| 11001001 | -1.0       | 11001101 | -1.0       | Others   | 0.0        |          |            |

Table 22: Euler, Length: 54, Numbers: Number 0.0: 201, Number -1.0: 31, Number 1.0: 20, Number -2.0: 3

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | 1.0        | 00000010 | -1.0       | 00000110 | -1.0       | 00000111 | -1.0       |
| 00010010 | -2.0       | 00010011 | -1.0       | 00010110 | -1.0       | 00010111 | -1.0       |
| 00011000 | -1.0       | 00011001 | -1.0       | 00011010 | -1.0       | 00011011 | -1.0       |
| 00100000 | 1.0        | 00100001 | 1.0        | 00100011 | 1.0        | 00110010 | -1.0       |
| 00111000 | -1.0       | 00111001 | -1.0       | 00111010 | -1.0       | 00111011 | -1.0       |
| 01000010 | -2.0       | 01000011 | -1.0       | 01000110 | -1.0       | 01000111 | -1.0       |
| 01010010 | -2.0       | 01010011 | -1.0       | 01010110 | -1.0       | 01010111 | -1.0       |
| 01100010 | -1.0       | 01110010 | -1.0       | 10000010 | -1.0       | 10010010 | -1.0       |
| 10010100 | 1.0        | 10010101 | 1.0        | 10010110 | 1.0        | 10010111 | 1.0        |
| 10011100 | 1.0        | 10011101 | 1.0        | 10011110 | 1.0        | 10011111 | 1.0        |
| 10100010 | -1.0       | 10110010 | -1.0       | 10110100 | 1.0        | 10110101 | 1.0        |
| 10110110 | 1.0        | 10110111 | 1.0        | 10111100 | 1.0        | 10111101 | 1.0        |
| 10111110 | 1.0        | 10111111 | 1.0        | 11000010 | -1.0       | 11010010 | -1.0       |
| 11100010 | -1.0       | 11110010 | -1.0       | Others   | 0.0        |          |            |

## 1.5 Tables from Euler using OMP Regression model with 80 coefficients

Table 23: Euler, Length: 80, Numbers: Number 0.0: 175, Number -0.5: 33, Number 0.5: 31, Number -1.0: 16

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 0.5        | 00000100 | 0.5        | 00001001 | -0.5       | 00001011 | -0.5       |
| 00011000 | -1.0       | 00011001 | -0.5       | 00011010 | -0.5       | 00011011 | -0.5       |
| 00011100 | -0.5       | 00100000 | 0.5        | 00100010 | 0.5        | 00101011 | -0.5       |
| 00101101 | 0.5        | 00111000 | -0.5       | 00111010 | -0.5       | 00111011 | -0.5       |
| 00111101 | 0.5        | 01000000 | -0.5       | 01000001 | -1.0       | 01000010 | -1.0       |
| 01000011 | -1.0       | 01000100 | -0.5       | 01000101 | -1.0       | 01000110 | -1.0       |
| 01000111 | -1.0       | 01001000 | -1.0       | 01001001 | -1.0       | 01001010 | -1.0       |
| 01001011 | -1.0       | 01001100 | -1.0       | 01001101 | -1.0       | 01001110 | -1.0       |
| 01001111 | -1.0       | 01011000 | -0.5       | 01011100 | -0.5       | 01101001 | 0.5        |
| 01101101 | 0.5        | 01111001 | 0.5        | 01111101 | 0.5        | 10000001 | -1.0       |
| 10000011 | -0.5       | 10000101 | -0.5       | 10001001 | -0.5       | 10001011 | -0.5       |
| 10010000 | -0.5       | 10010001 | -0.5       | 10010110 | 0.5        | 10010111 | 0.5        |
| 10011000 | -0.5       | 10011101 | 0.5        | 10011110 | 0.5        | 10011111 | 0.5        |
| 10100001 | -0.5       | 10100011 | -0.5       | 10101011 | -0.5       | 10101101 | 0.5        |
| 10110000 | -0.5       | 10110001 | -0.5       | 10110010 | -0.5       | 10110011 | -0.5       |
| 10111000 | -0.5       | 10111010 | -0.5       | 10111011 | -0.5       | 10111101 | 0.5        |
| 11000001 | -0.5       | 11000101 | -0.5       | 11010010 | 0.5        | 11010011 | 0.5        |
| 11010110 | 0.5        | 11010111 | 0.5        | 11011001 | 0.5        | 11011010 | 0.5        |
| 11011011 | 0.5        | 11011101 | 0.5        | 11011110 | 0.5        | 11011111 | 0.5        |
| 11101001 | 0.5        | 11101101 | 0.5        | 11111001 | 0.5        | 11111101 | 0.5        |
| Others   | 0.0        |          |            |          |            |          |            |

Table 24: Euler, Length: 80, Numbers: Number 0.0: 175, Number -0.5: 33, Number 0.5: 31, Number -1.0: 16

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | 0.5        | 00000010 | 0.5        | 00000011 | 0.5        | 00010100 | -0.5       |
| 00010101 | -0.5       | 00010111 | -0.5       | 00011000 | -1.0       | 00011001 | -0.5       |
| 00011010 | -0.5       | 00011011 | -0.5       | 00011100 | -0.5       | 00011101 | -0.5       |
| 00011111 | -0.5       | 00100100 | -1.0       | 00100101 | -0.5       | 00100110 | -0.5       |
| 00100111 | -0.5       | 00101000 | -0.5       | 00101010 | -0.5       | 00101011 | -0.5       |
| 00101100 | -0.5       | 00101110 | -0.5       | 00101111 | -0.5       | 00110100 | -0.5       |
| 00110101 | -0.5       | 00110111 | -0.5       | 00111000 | -0.5       | 00111010 | -0.5       |
| 00111011 | -0.5       | 00111111 | -0.5       | 01000000 | -0.5       | 01000010 | -1.0       |
| 01001000 | -1.0       | 01001010 | -1.0       | 01010110 | 0.5        | 01011000 | -0.5       |
| 01011110 | 0.5        | 01100000 | -1.0       | 01100010 | -1.0       | 01100100 | -0.5       |
| 01101000 | -1.0       | 01101001 | 0.5        | 01101010 | -1.0       | 01101101 | 0.5        |
| 01110110 | 0.5        | 01111001 | 0.5        | 01111100 | 0.5        | 01111101 | 0.5        |
| 01111110 | 0.5        | 10000000 | 0.5        | 10010110 | 0.5        | 10011000 | -0.5       |
| 10011110 | 0.5        | 10100100 | -0.5       | 10101001 | 0.5        | 10101101 | 0.5        |
| 10110110 | 0.5        | 10111001 | 0.5        | 10111100 | 0.5        | 10111101 | 0.5        |
| 10111110 | 0.5        | 11000000 | -0.5       | 11000010 | -1.0       | 11001000 | -1.0       |
| 11001010 | -1.0       | 11010110 | 0.5        | 11011000 | -0.5       | 11011110 | 0.5        |
| 11100000 | -1.0       | 11100010 | -1.0       | 11100100 | -0.5       | 11101000 | -1.0       |
| 11101001 | 0.5        | 11101010 | -1.0       | 11101101 | 0.5        | 11110110 | 0.5        |
| 11111001 | 0.5        | 11111100 | 0.5        | 11111101 | 0.5        | 11111110 | 0.5        |
| Others   | 0.0        |          |            |          |            |          |            |

Table 25: Euler, Length: 80, Numbers: Number 0.0: 175, Number -0.5: 33, Number 0.5: 31, Number -1.0: 16

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | -0.5       | 00000110 | -0.5       | 00001000 | 0.5        | 00001110 | -0.5       |
| 00010000 | 0.5        | 00010001 | -0.5       | 00100001 | -1.0       | 00100100 | -1.0       |
| 00100101 | -0.5       | 00100110 | -0.5       | 00101100 | -0.5       | 00101110 | -0.5       |
| 00110001 | -1.0       | 00110100 | -0.5       | 00110101 | -0.5       | 01000001 | -1.0       |
| 01000010 | -1.0       | 01000011 | -0.5       | 01000110 | -0.5       | 01001010 | -0.5       |
| 01001110 | -0.5       | 01010001 | -1.0       | 01010010 | -0.5       | 01010011 | -0.5       |
| 01100000 | -0.5       | 01100001 | -1.0       | 01100010 | -0.5       | 01100100 | -0.5       |
| 01100111 | 0.5        | 01101001 | 0.5        | 01101011 | 0.5        | 01101101 | 0.5        |
| 01101111 | 0.5        | 01110001 | -1.0       | 01110110 | 0.5        | 01110111 | 0.5        |
| 01111000 | 0.5        | 01111001 | 0.5        | 01111010 | 0.5        | 01111011 | 0.5        |
| 01111100 | 0.5        | 01111101 | 0.5        | 01111110 | 0.5        | 01111111 | 0.5        |
| 10000000 | 0.5        | 10000001 | -1.0       | 10000111 | 0.5        | 10001000 | 0.5        |
| 10001110 | -0.5       | 10010001 | -1.0       | 10010110 | 0.5        | 10010111 | 0.5        |
| 10100001 | -1.0       | 10100100 | -0.5       | 10100111 | 0.5        | 10101100 | -0.5       |
| 10101110 | -0.5       | 10110001 | -1.0       | 10110110 | 0.5        | 10110111 | 0.5        |
| 11000001 | -1.0       | 11000010 | -0.5       | 11000111 | 0.5        | 11001010 | -0.5       |
| 11001110 | -0.5       | 11010001 | -1.0       | 11010110 | 0.5        | 11010111 | 0.5        |
| 11100000 | -0.5       | 11100001 | -1.0       | 11100010 | -0.5       | 11100100 | -0.5       |
| 11100111 | 0.5        | 11101000 | -0.5       | 11101010 | -0.5       | 11101100 | -0.5       |
| 11101110 | -0.5       | 11110001 | -1.0       | 11110110 | 0.5        | 11110111 | 0.5        |
| Others   | 0.0        |          |            |          |            |          |            |



Table 26: Euler, Length: 80, Numbers: Number 0.0: 175, Number -0.5: 33, Number 0.5: 31, Number -1.0: 16

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 0.5        | 00000100 | 0.5        | 00001001 | -0.5       | 00001101 | -0.5       |
| 00011000 | -1.0       | 00011001 | -0.5       | 00011010 | -0.5       | 00011100 | -0.5       |
| 00011101 | -0.5       | 00100000 | -0.5       | 00100001 | -1.0       | 00100010 | -0.5       |
| 00100011 | -1.0       | 00100100 | -1.0       | 00100101 | -1.0       | 00100110 | -1.0       |
| 00100111 | -1.0       | 00101000 | -1.0       | 00101001 | -1.0       | 00101010 | -1.0       |
| 00101011 | -1.0       | 00101100 | -1.0       | 00101101 | -1.0       | 00101110 | -1.0       |
| 00101111 | -1.0       | 00111000 | -0.5       | 00111010 | -0.5       | 01000000 | 0.5        |
| 01000100 | 0.5        | 01001011 | 0.5        | 01001101 | -0.5       | 01011000 | -0.5       |
| 01011011 | 0.5        | 01011100 | -0.5       | 01011101 | -0.5       | 01101001 | 0.5        |
| 01101011 | 0.5        | 01111001 | 0.5        | 01111011 | 0.5        | 10000001 | -1.0       |
| 10000011 | -0.5       | 10000101 | -0.5       | 10001001 | -0.5       | 10001101 | -0.5       |
| 10010000 | -0.5       | 10010001 | -0.5       | 10010110 | 0.5        | 10010111 | 0.5        |
| 10011000 | -0.5       | 10011011 | 0.5        | 10011110 | 0.5        | 10011111 | 0.5        |
| 10100001 | -0.5       | 10100011 | -0.5       | 10110100 | 0.5        | 10110101 | 0.5        |
| 10110110 | 0.5        | 10110111 | 0.5        | 10111001 | 0.5        | 10111011 | 0.5        |
| 10111100 | 0.5        | 10111101 | 0.5        | 10111110 | 0.5        | 10111111 | 0.5        |
| 11000001 | -0.5       | 11000101 | -0.5       | 11001011 | 0.5        | 11001101 | -0.5       |
| 11010000 | -0.5       | 11010001 | -0.5       | 11010100 | -0.5       | 11010101 | -0.5       |
| 11011000 | -0.5       | 11011011 | 0.5        | 11011100 | -0.5       | 11011101 | -0.5       |
| 11101001 | 0.5        | 11101011 | 0.5        | 11111001 | 0.5        | 11111011 | 0.5        |
| Others   | 0.0        |          |            |          |            |          |            |

## 1.6 Tables from Enclosing Surface using OMP Regression model with 103 coefficients

Table 27: Enclosing\_Surface, Length: 103, Numbers: Number 0.0: 152, Number 6.0: 30, Number 2.0: 26, Number 4.0: 26, Number -2.0: 13, Number 8.0: 4, Number 10.0: 3, Number 12.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000101 | 2.0        | 00001000 | -2.0       | 00001100 | -2.0       | 00010000 | 2.0        |
| 00010001 | 6.0        | 00010010 | 2.0        | 00010011 | 6.0        | 00010100 | 6.0        |
| 00010101 | 8.0        | 00010110 | 6.0        | 00010111 | 6.0        | 00011001 | 6.0        |
| 00011010 | 2.0        | 00011011 | 6.0        | 00011100 | 4.0        | 00011101 | 6.0        |
| 00011110 | 6.0        | 00011111 | 6.0        | 00100000 | 6.0        | 00100001 | 4.0        |
| 00100100 | 6.0        | 00100101 | 6.0        | 00110000 | 6.0        | 00110001 | 10.0       |
| 00110011 | 4.0        | 00110100 | 10.0       | 00110101 | 12.0       | 00110110 | 4.0        |
| 00110111 | 4.0        | 00111001 | 6.0        | 00111011 | 4.0        | 00111100 | 4.0        |
| 00111101 | 6.0        | 00111110 | 4.0        | 00111111 | 4.0        | 01000101 | 2.0        |
| 01001000 | -2.0       | 01001100 | -2.0       | 01010101 | 2.0        | 01011000 | -2.0       |
| 01011100 | -2.0       | 01100000 | 6.0        | 01100001 | 4.0        | 01100100 | 6.0        |
| 01100101 | 6.0        | 01110000 | 6.0        | 01110001 | 6.0        | 01110100 | 6.0        |
| 01110101 | 8.0        | 01111001 | 2.0        | 01111101 | 2.0        | 10000100 | 2.0        |
| 10000101 | 4.0        | 10000110 | 2.0        | 10000111 | 2.0        | 10001000 | -2.0       |
| 10001100 | -2.0       | 10010001 | 4.0        | 10010011 | 4.0        | 10010100 | 6.0        |
| 10010101 | 8.0        | 10010110 | 6.0        | 10010111 | 6.0        | 10011000 | -2.0       |
| 10011001 | 4.0        | 10011011 | 4.0        | 10011100 | 2.0        | 10011101 | 4.0        |
| 10011110 | 4.0        | 10011111 | 4.0        | 10100000 | 2.0        | 10100100 | 4.0        |
| 10100101 | 4.0        | 10100110 | 2.0        | 10100111 | 2.0        | 10110000 | 2.0        |
| 10110001 | 6.0        | 10110011 | 4.0        | 10110100 | 8.0        | 10110101 | 10.0       |
| 10110110 | 6.0        | 10110111 | 6.0        | 10111001 | 6.0        | 10111011 | 4.0        |
| 10111100 | 4.0        | 10111101 | 6.0        | 10111110 | 4.0        | 10111111 | 4.0        |
| 11000101 | 2.0        | 11001000 | -2.0       | 11001100 | -2.0       | 11010101 | 2.0        |
| 11011000 | -2.0       | 11011100 | -2.0       | 11100000 | 2.0        | 11100100 | 2.0        |
| 11100101 | 2.0        | 11110000 | 2.0        | 11110001 | 2.0        | 11110100 | 2.0        |
| 11110101 | 4.0        | 11111001 | 2.0        | 11111101 | 2.0        | Others   | 0.0        |

Table 28: Enclosing\_Surface, Length: 103, Numbers: Number 0.0: 152, Number 6.0: 30, Number 2.0: 26, Number 4.0: 26, Number -2.0: 13, Number 8.0: 4, Number 10.0: 3, Number 12.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 6.0        | 00000011 | 2.0        | 00000110 | 6.0        | 00000111 | 2.0        |
| 00001000 | 2.0        | 00001010 | 6.0        | 00001011 | 2.0        | 00001110 | 6.0        |
| 00001111 | 2.0        | 00010000 | -2.0       | 00010001 | -2.0       | 00010100 | -2.0       |
| 00010101 | -2.0       | 00011001 | -2.0       | 00011100 | -2.0       | 00011101 | -2.0       |
| 00101000 | 2.0        | 00111000 | 2.0        | 01000001 | 2.0        | 01000010 | 6.0        |
| 01000011 | 4.0        | 01000110 | 6.0        | 01000111 | 2.0        | 01001000 | 6.0        |
| 01001001 | 6.0        | 01001010 | 10.0       | 01001011 | 8.0        | 01001110 | 6.0        |
| 01001111 | 2.0        | 01010000 | -2.0       | 01010001 | -2.0       | 01010100 | -2.0       |
| 01010101 | -2.0       | 01011000 | 4.0        | 01011001 | 2.0        | 01011010 | 4.0        |
| 01011011 | 4.0        | 01011100 | -2.0       | 01011101 | -2.0       | 01100001 | 2.0        |
| 01100011 | 2.0        | 01101000 | 6.0        | 01101001 | 6.0        | 01101010 | 4.0        |
| 01101011 | 6.0        | 01111000 | 6.0        | 01111001 | 4.0        | 01111010 | 4.0        |
| 01111011 | 4.0        | 10000010 | 4.0        | 10000110 | 4.0        | 10001000 | 6.0        |
| 10001001 | 4.0        | 10001010 | 10.0       | 10001011 | 6.0        | 10001110 | 6.0        |
| 10001111 | 2.0        | 10011000 | 6.0        | 10011001 | 4.0        | 10011010 | 6.0        |
| 10011011 | 6.0        | 10011110 | 2.0        | 10011111 | 2.0        | 10101000 | 6.0        |
| 10101001 | 4.0        | 10101010 | 4.0        | 10101011 | 4.0        | 10111000 | 6.0        |
| 10111001 | 4.0        | 10111010 | 4.0        | 10111011 | 4.0        | 11000000 | 2.0        |
| 11000001 | 4.0        | 11000010 | 6.0        | 11000011 | 4.0        | 11000100 | 2.0        |
| 11000101 | 2.0        | 11000110 | 6.0        | 11000111 | 2.0        | 11001000 | 8.0        |
| 11001001 | 8.0        | 11001010 | 12.0       | 11001011 | 10.0       | 11001100 | 2.0        |
| 11001101 | 2.0        | 11001110 | 8.0        | 11001111 | 4.0        | 11011000 | 6.0        |
| 11011001 | 4.0        | 11011010 | 6.0        | 11011011 | 6.0        | 11011110 | 2.0        |
| 11011111 | 2.0        | 11100001 | 2.0        | 11100011 | 2.0        | 11101000 | 6.0        |
| 11101001 | 6.0        | 11101010 | 4.0        | 11101011 | 6.0        | 11111000 | 6.0        |
| 11111001 | 4.0        | 11111010 | 4.0        | 11111011 | 4.0        | Others   | 0.0        |

## 1.7 Tables from Enclosing Surface using OMP Regression model with 105 coefficients

Table 29: Enclosing\_Surface, Length: 105, Numbers: Number 0.0: 150, Number 2.0: 33, Number 6.0: 28, Number 4.0: 23, Number -2.0: 12, Number 8.0: 5, Number 10.0: 3, Number 12.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | -2.0       | 00000011 | -2.0       | 00000100 | 6.0        | 00000110 | 4.0        |
| 00010010 | -2.0       | 00010011 | -2.0       | 00010100 | 6.0        | 00010110 | 4.0        |
| 00100010 | -2.0       | 00100011 | -2.0       | 00100100 | 6.0        | 00100110 | 4.0        |
| 00110010 | -2.0       | 00110011 | -2.0       | 00110100 | 6.0        | 00110110 | 4.0        |
| 01000000 | 2.0        | 01000011 | -2.0       | 01000100 | 6.0        | 01000110 | 4.0        |
| 01001000 | 2.0        | 01001010 | 2.0        | 01010000 | 2.0        | 01010001 | 2.0        |
| 01010100 | 8.0        | 01010101 | 2.0        | 01010110 | 6.0        | 01010111 | 2.0        |
| 01011000 | 2.0        | 01011001 | 2.0        | 01011010 | 2.0        | 01011011 | 2.0        |
| 01011100 | 2.0        | 01011101 | 2.0        | 01011110 | 2.0        | 01011111 | 2.0        |
| 01100000 | 2.0        | 01100011 | -2.0       | 01100100 | 6.0        | 01100110 | 4.0        |
| 01101000 | 2.0        | 01101010 | 2.0        | 01110010 | -2.0       | 01110011 | -2.0       |
| 01110100 | 6.0        | 01110110 | 4.0        | 10000100 | 6.0        | 10000110 | 6.0        |
| 10000111 | 2.0        | 10010100 | 6.0        | 10010110 | 6.0        | 10010111 | 2.0        |
| 10100000 | 2.0        | 10100001 | 2.0        | 10100100 | 8.0        | 10100101 | 2.0        |
| 10100110 | 6.0        | 10100111 | 2.0        | 10110000 | 2.0        | 10110001 | 2.0        |
| 10110100 | 8.0        | 10110101 | 2.0        | 10110110 | 6.0        | 10110111 | 2.0        |
| 11000000 | 6.0        | 11000001 | 4.0        | 11000010 | 6.0        | 11000011 | 4.0        |
| 11000100 | 10.0       | 11000101 | 4.0        | 11000110 | 10.0       | 11000111 | 6.0        |
| 11001000 | 6.0        | 11001001 | 4.0        | 11001010 | 6.0        | 11001011 | 4.0        |
| 11001100 | 4.0        | 11001101 | 4.0        | 11001110 | 4.0        | 11001111 | 4.0        |
| 11010100 | 6.0        | 11010110 | 6.0        | 11010111 | 2.0        | 11100000 | 8.0        |
| 11100001 | 6.0        | 11100010 | 6.0        | 11100011 | 4.0        | 11100100 | 12.0       |
| 11100101 | 6.0        | 11100110 | 10.0       | 11100111 | 6.0        | 11101000 | 6.0        |
| 11101001 | 4.0        | 11101010 | 6.0        | 11101011 | 4.0        | 11101100 | 4.0        |
| 11101101 | 4.0        | 11101110 | 4.0        | 11101111 | 4.0        | 11110000 | 2.0        |
| 11110001 | 2.0        | 11110100 | 8.0        | 11110101 | 2.0        | 11110110 | 6.0        |
| 11110111 | 2.0        | Others   | 0.0        |          |            |          |            |

Table 30: Enclosing\_Surface, Length: 105, Numbers: Number 0.0: 150, Number 2.0: 31, Number 4.0: 29, Number 6.0: 26, Number -2.0: 11, Number 8.0: 5, Number 10.0: 2, Number 12.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001000 | 4.0        | 00001001 | 4.0        | 00001010 | 4.0        | 00001011 | 6.0        |
| 00010000 | 2.0        | 00010010 | 2.0        | 00011000 | 6.0        | 00011001 | 4.0        |
| 00011010 | 6.0        | 00011011 | 6.0        | 00101000 | 6.0        | 00101001 | 6.0        |
| 00101010 | 4.0        | 00101011 | 6.0        | 00101100 | 2.0        | 00101101 | 2.0        |
| 00110000 | 4.0        | 00110001 | 2.0        | 00110010 | 8.0        | 00110011 | 2.0        |
| 00110100 | 2.0        | 00110101 | 2.0        | 00110110 | 6.0        | 00110111 | 2.0        |
| 00111000 | 10.0       | 00111001 | 8.0        | 00111010 | 12.0       | 00111011 | 8.0        |
| 00111100 | 4.0        | 00111101 | 4.0        | 00111110 | 6.0        | 00111111 | 2.0        |
| 01001000 | 4.0        | 01001001 | 4.0        | 01001010 | 4.0        | 01001011 | 6.0        |
| 01010001 | -2.0       | 01010011 | -2.0       | 01011000 | 4.0        | 01011001 | 2.0        |
| 01011010 | 4.0        | 01011011 | 4.0        | 01101000 | 6.0        | 01101001 | 6.0        |
| 01101010 | 4.0        | 01101011 | 6.0        | 01101100 | 2.0        | 01101101 | 2.0        |
| 01110000 | 2.0        | 01110010 | 6.0        | 01110100 | 2.0        | 01110101 | 2.0        |
| 01110110 | 6.0        | 01110111 | 2.0        | 01111000 | 8.0        | 01111001 | 6.0        |
| 01111010 | 10.0       | 01111011 | 6.0        | 01111100 | 4.0        | 01111101 | 4.0        |
| 01111110 | 6.0        | 01111111 | 2.0        | 10001000 | 4.0        | 10001001 | 4.0        |
| 10001010 | 4.0        | 10001011 | 6.0        | 10010000 | 2.0        | 10010010 | 2.0        |
| 10011000 | 6.0        | 10011001 | 4.0        | 10011010 | 6.0        | 10011011 | 6.0        |
| 10101000 | 4.0        | 10101001 | 4.0        | 10101010 | 2.0        | 10101011 | 4.0        |
| 10101110 | -2.0       | 10101111 | -2.0       | 10110000 | 2.0        | 10110010 | 6.0        |
| 10110110 | 4.0        | 10111000 | 6.0        | 10111001 | 4.0        | 10111010 | 8.0        |
| 10111011 | 4.0        | 10111110 | 2.0        | 10111111 | -2.0       | 11001011 | 2.0        |
| 11010001 | -2.0       | 11010011 | -2.0       | 11011001 | -2.0       | 11101010 | -2.0       |
| 11101110 | -2.0       | 11101111 | -2.0       | 11110000 | 2.0        | 11110010 | 6.0        |
| 11110100 | 2.0        | 11110101 | 2.0        | 11110110 | 6.0        | 11110111 | 2.0        |
| 11111000 | 2.0        | 11111010 | 4.0        | 11111100 | 2.0        | 11111101 | 2.0        |
| 11111110 | 4.0        | Others   | 0.0        |          |            |          |            |

## 1.8 Tables from Contact Surface using OMP Regression model with 104 coefficients

Table 31: Contact\_Surface, Length: 104, Numbers: Number 0.0: 151, Number 1.0: 36, Number 2.0: 26, Number 3.0: 24, Number -1.0: 9, Number 4.0: 6, Number 6.0: 2, Number 5.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 1.0        | 00000110 | 1.0        | 00001010 | 1.0        | 00010010 | 1.0        |
| 00010110 | 1.0        | 00011010 | 1.0        | 00100010 | 1.0        | 00100110 | 1.0        |
| 00101010 | 1.0        | 00110010 | 3.0        | 00110110 | 3.0        | 00111010 | 3.0        |
| 00111110 | 2.0        | 01000000 | -1.0       | 01000110 | 1.0        | 01001000 | -1.0       |
| 01010010 | 1.0        | 01010100 | 1.0        | 01010110 | 2.0        | 01011010 | 1.0        |
| 01011100 | 1.0        | 01011110 | 1.0        | 01100000 | -1.0       | 01100110 | 1.0        |
| 01101000 | -1.0       | 01110010 | 3.0        | 01110100 | 1.0        | 01110110 | 4.0        |
| 01111010 | 3.0        | 01111100 | 1.0        | 01111110 | 3.0        | 10000011 | -1.0       |
| 10000111 | -1.0       | 10001010 | 1.0        | 10010011 | -1.0       | 10010111 | -1.0       |
| 10011010 | 1.0        | 10100010 | 1.0        | 10100110 | 1.0        | 10101010 | 2.0        |
| 10101011 | 1.0        | 10101110 | 1.0        | 10101111 | 1.0        | 10110010 | 3.0        |
| 10110110 | 3.0        | 10111010 | 4.0        | 10111011 | 1.0        | 10111110 | 3.0        |
| 10111111 | 1.0        | 11000000 | 1.0        | 11000001 | 2.0        | 11000010 | 1.0        |
| 11000011 | 1.0        | 11000100 | 2.0        | 11000101 | 2.0        | 11000110 | 2.0        |
| 11000111 | 1.0        | 11001000 | -1.0       | 11001100 | 2.0        | 11001101 | 2.0        |
| 11001110 | 2.0        | 11001111 | 2.0        | 11010000 | 2.0        | 11010001 | 2.0        |
| 11010010 | 2.0        | 11010011 | 1.0        | 11010100 | 3.0        | 11010101 | 2.0        |
| 11010110 | 3.0        | 11010111 | 1.0        | 11011010 | 1.0        | 11011100 | 3.0        |
| 11011101 | 2.0        | 11011110 | 3.0        | 11011111 | 2.0        | 11100000 | 2.0        |
| 11100001 | 3.0        | 11100010 | 3.0        | 11100011 | 3.0        | 11100100 | 3.0        |
| 11100101 | 3.0        | 11100110 | 4.0        | 11100111 | 3.0        | 11101001 | 1.0        |
| 11101010 | 2.0        | 11101011 | 2.0        | 11101100 | 3.0        | 11101101 | 3.0        |
| 11101110 | 4.0        | 11101111 | 4.0        | 11110000 | 2.0        | 11110001 | 2.0        |
| 11110010 | 5.0        | 11110011 | 2.0        | 11110100 | 3.0        | 11110101 | 2.0        |
| 11110110 | 6.0        | 11110111 | 2.0        | 11111010 | 4.0        | 11111011 | 1.0        |
| 11111100 | 3.0        | 11111101 | 2.0        | 11111110 | 6.0        | 11111111 | 3.0        |
| Others   | 0.0        |          |            |          |            |          |            |

Table 32: Contact\_Surface, Length: 104, Numbers: Number 0.0: 151, Number 1.0: 30, Number 2.0: 29, Number 3.0: 21, Number -1.0: 12, Number 4.0: 9, Number 6.0: 2, Number 5.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000011 | 1.0        | 00000101 | 1.0        | 00000111 | 2.0        | 00001011 | 1.0        |
| 00001101 | 1.0        | 00001111 | 2.0        | 00010001 | -1.0       | 00010111 | 1.0        |
| 00011001 | -1.0       | 00011111 | 1.0        | 00100101 | 1.0        | 00100111 | 1.0        |
| 00101101 | 1.0        | 00101111 | 1.0        | 00110001 | -1.0       | 00110111 | 1.0        |
| 00111001 | -1.0       | 00111111 | 1.0        | 01000011 | 1.0        | 01000111 | 1.0        |
| 01001011 | 1.0        | 01001111 | 1.0        | 01010001 | -1.0       | 01010101 | 2.0        |
| 01010111 | 3.0        | 01011001 | -1.0       | 01011101 | 2.0        | 01011111 | 3.0        |
| 01110001 | -1.0       | 01110101 | 2.0        | 01110111 | 3.0        | 01111001 | -1.0       |
| 01111101 | 2.0        | 01111111 | 3.0        | 10000011 | 1.0        | 10000101 | 1.0        |
| 10000111 | 2.0        | 10001000 | 2.0        | 10001001 | 2.0        | 10001010 | 3.0        |
| 10001011 | 4.0        | 10001100 | 3.0        | 10001101 | 4.0        | 10001110 | 4.0        |
| 10001111 | 6.0        | 10010001 | -1.0       | 10010111 | 1.0        | 10011000 | 2.0        |
| 10011001 | 1.0        | 10011010 | 3.0        | 10011011 | 3.0        | 10011100 | 3.0        |
| 10011101 | 3.0        | 10011110 | 4.0        | 10011111 | 5.0        | 10100101 | 1.0        |
| 10100111 | 1.0        | 10101000 | 2.0        | 10101001 | 2.0        | 10101100 | 3.0        |
| 10101101 | 4.0        | 10101110 | 1.0        | 10101111 | 2.0        | 10110001 | -1.0       |
| 10110111 | 1.0        | 10111000 | 2.0        | 10111001 | 1.0        | 10111100 | 3.0        |
| 10111101 | 3.0        | 10111110 | 1.0        | 10111111 | 2.0        | 11000011 | 1.0        |
| 11000111 | 1.0        | 11001000 | 2.0        | 11001001 | 2.0        | 11001010 | 3.0        |
| 11001011 | 4.0        | 11001100 | 2.0        | 11001101 | 2.0        | 11001110 | 3.0        |
| 11001111 | 4.0        | 11010001 | -1.0       | 11010101 | 2.0        | 11010111 | 3.0        |
| 11011000 | 2.0        | 11011001 | 1.0        | 11011010 | 3.0        | 11011011 | 3.0        |
| 11011100 | 2.0        | 11011101 | 4.0        | 11011110 | 3.0        | 11011111 | 6.0        |
| 11101000 | 2.0        | 11101001 | 2.0        | 11101100 | 2.0        | 11101101 | 2.0        |
| 11110001 | -1.0       | 11110101 | 2.0        | 11110111 | 3.0        | 11111000 | 2.0        |
| 11111001 | 1.0        | 11111100 | 2.0        | 11111101 | 4.0        | 11111111 | 3.0        |
| Others   | 0.0        |          |            |          |            |          |            |

## 1.9 Tables from Contact Surface using OMP Regression model with 109 coefficients

Table 33: Contact\_Surface, Length: 109, Numbers: Number 0.0: 146, Number 2.0: 30, Number 1.0: 23, Number -1.0: 16, Number 3.0: 16, Number 4.0: 8, Number 6.0: 6, Number -2.0: 4, Number 5.0: 3, Number -3.0: 2, Number 8.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | -1.0       | 00000011 | -1.0       | 00000110 | -1.0       | 00001100 | 2.0        |
| 00001101 | 2.0        | 00001110 | 2.0        | 00001111 | 2.0        | 00010001 | 1.0        |
| 00010011 | 2.0        | 00010101 | 4.0        | 00010111 | 6.0        | 00011001 | 1.0        |
| 00011010 | 1.0        | 00011011 | 3.0        | 00011100 | 2.0        | 00011101 | 6.0        |
| 00011110 | 3.0        | 00011111 | 8.0        | 00100011 | 2.0        | 00100111 | 3.0        |
| 00101010 | -2.0       | 00101100 | 2.0        | 00101101 | 2.0        | 00101111 | 2.0        |
| 00110011 | 2.0        | 00110101 | 3.0        | 00110111 | 6.0        | 00111010 | -2.0       |
| 00111100 | 2.0        | 00111101 | 5.0        | 00111111 | 5.0        | 01000000 | 1.0        |
| 01000001 | 1.0        | 01000110 | -1.0       | 01010000 | 1.0        | 01010001 | 2.0        |
| 01010010 | 1.0        | 01010011 | 3.0        | 01010101 | 2.0        | 01010111 | 4.0        |
| 01011001 | 1.0        | 01011010 | 1.0        | 01011011 | 3.0        | 01011101 | 2.0        |
| 01011110 | 1.0        | 01011111 | 4.0        | 01100000 | 1.0        | 01100001 | 1.0        |
| 01100010 | 1.0        | 01100011 | 3.0        | 01100111 | 3.0        | 01101010 | -2.0       |
| 01101110 | -2.0       | 01110011 | 2.0        | 01110100 | -1.0       | 01110110 | -1.0       |
| 01110111 | 3.0        | 01111000 | -1.0       | 01111001 | -1.0       | 01111010 | -3.0       |
| 01111011 | -1.0       | 01111100 | -1.0       | 01111110 | -3.0       | 10000010 | -1.0       |
| 10000011 | -1.0       | 10000110 | -1.0       | 10010001 | 1.0        | 10010011 | 2.0        |
| 10010101 | 4.0        | 10010111 | 6.0        | 10011001 | 1.0        | 10011010 | 1.0        |
| 10011011 | 3.0        | 10011101 | 4.0        | 10011110 | 1.0        | 10011111 | 6.0        |
| 10100011 | 2.0        | 10100111 | 3.0        | 10101011 | 2.0        | 10101111 | 2.0        |
| 10110011 | 2.0        | 10110101 | 3.0        | 10110111 | 6.0        | 10111011 | 2.0        |
| 10111101 | 3.0        | 10111111 | 5.0        | 11000010 | -1.0       | 11000011 | -1.0       |
| 11000110 | -1.0       | 11010001 | 1.0        | 11010011 | 2.0        | 11010101 | 2.0        |
| 11010111 | 4.0        | 11011001 | 1.0        | 11011010 | 1.0        | 11011011 | 3.0        |
| 11011101 | 2.0        | 11011110 | 1.0        | 11011111 | 4.0        | 11100011 | 2.0        |
| 11100111 | 3.0        | 11101011 | 2.0        | 11101111 | 2.0        | 11110011 | 2.0        |
| 11110101 | 1.0        | 11110111 | 4.0        | 11111011 | 2.0        | 11111101 | 1.0        |
| 11111111 | 3.0        | Others   | 0.0        |          |            |          |            |



Table 34: Contact\_Surface, Length: 109, Numbers: Number 0.0: 146, Number 2.0: 30, Number 1.0: 24, Number -1.0: 14, Number 3.0: 11, Number 4.0: 10, Number 6.0: 6, Number -3.0: 5, Number -2.0: 4, Number 8.0: 3, Number 7.0: 1, Number 5.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | -1.0       | 00000101 | -3.0       | 00001010 | 2.0        | 00001011 | 2.0        |
| 00001110 | 2.0        | 00001111 | 2.0        | 00010011 | 4.0        | 00010111 | 6.0        |
| 00011001 | 1.0        | 00011010 | 2.0        | 00011011 | 6.0        | 00011101 | 3.0        |
| 00011110 | 2.0        | 00011111 | 8.0        | 00100001 | -1.0       | 00100101 | -3.0       |
| 00110111 | 2.0        | 00111001 | 1.0        | 00111101 | 3.0        | 00111111 | 2.0        |
| 01000001 | -1.0       | 01000100 | 1.0        | 01000110 | 1.0        | 01000111 | 3.0        |
| 01001010 | 2.0        | 01001011 | 2.0        | 01001100 | -2.0       | 01001111 | 2.0        |
| 01010000 | 2.0        | 01010001 | 2.0        | 01010010 | 2.0        | 01010011 | 6.0        |
| 01010100 | 2.0        | 01010101 | 2.0        | 01010110 | 2.0        | 01010111 | 8.0        |
| 01011000 | 2.0        | 01011001 | 3.0        | 01011010 | 4.0        | 01011011 | 8.0        |
| 01011100 | -1.0       | 01011101 | 2.0        | 01011110 | 1.0        | 01011111 | 7.0        |
| 01100001 | -1.0       | 01100100 | 1.0        | 01100110 | 1.0        | 01100111 | 3.0        |
| 01101100 | -2.0       | 01101110 | -2.0       | 01110111 | 2.0        | 01111001 | 1.0        |
| 01111100 | -3.0       | 01111110 | -3.0       | 01111111 | -1.0       | 10000000 | 1.0        |
| 10000010 | 1.0        | 10000011 | 1.0        | 10000100 | 1.0        | 10000101 | -2.0       |
| 10000110 | 1.0        | 10000111 | 1.0        | 10010011 | 4.0        | 10010111 | 6.0        |
| 10011000 | -1.0       | 10011010 | -1.0       | 10011011 | 3.0        | 10011100 | -1.0       |
| 10011101 | 2.0        | 10011110 | -1.0       | 10011111 | 5.0        | 10100001 | -1.0       |
| 10100101 | -3.0       | 10110111 | 2.0        | 10111001 | 1.0        | 10111101 | 3.0        |
| 10111111 | 2.0        | 11000000 | 1.0        | 11000010 | 1.0        | 11000011 | 1.0        |
| 11000100 | 2.0        | 11000101 | 1.0        | 11000110 | 2.0        | 11000111 | 4.0        |
| 11001100 | 2.0        | 11001101 | 4.0        | 11001110 | 2.0        | 11001111 | 4.0        |
| 11010011 | 4.0        | 11010111 | 6.0        | 11011000 | -1.0       | 11011010 | -1.0       |
| 11011011 | 3.0        | 11011101 | 3.0        | 11011111 | 6.0        | 11100001 | -1.0       |
| 11100100 | 1.0        | 11100110 | 1.0        | 11100111 | 3.0        | 11101100 | 2.0        |
| 11101101 | 4.0        | 11101110 | 2.0        | 11101111 | 4.0        | 11110111 | 2.0        |
| 11111001 | 1.0        | 11111100 | 1.0        | 11111101 | 4.0        | 11111110 | 1.0        |
| 11111111 | 3.0        | Others   | 0.0        |          |            |          |            |

### 1.10 Tables from Volume using OMP Regression model with 93 coefficients

Table 35: Volume, Length: 93, Numbers: Number 0.0: 162, Number 1.0: 58, Number 2.0: 24, Number 3.0: 7, Number -1.0: 3, Number 4.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00100010 | 1.0        | 00100011 | 1.0        | 00100110 | 1.0        | 00100111 | 1.0        |
| 00101010 | 1.0        | 00101011 | 1.0        | 00101110 | 1.0        | 00101111 | 1.0        |
| 00110010 | 2.0        | 00110011 | 1.0        | 00110110 | 2.0        | 00110111 | 1.0        |
| 00111010 | 2.0        | 00111011 | 1.0        | 00111110 | 2.0        | 00111111 | 1.0        |
| 01000000 | 1.0        | 01000001 | 1.0        | 01000010 | 1.0        | 01000011 | 1.0        |
| 01001000 | 1.0        | 01001001 | 1.0        | 01001010 | 1.0        | 01001011 | 1.0        |
| 01010000 | 1.0        | 01010010 | 1.0        | 01011000 | 1.0        | 01011010 | 1.0        |
| 01100000 | 1.0        | 01100001 | 1.0        | 01100010 | 2.0        | 01100011 | 2.0        |
| 01100110 | 1.0        | 01100111 | 1.0        | 01101000 | 1.0        | 01101001 | 1.0        |
| 01101010 | 2.0        | 01101011 | 2.0        | 01101110 | 1.0        | 01101111 | 1.0        |
| 01110000 | 1.0        | 01110010 | 3.0        | 01110011 | 1.0        | 01110110 | 2.0        |
| 01110111 | 1.0        | 01111000 | 1.0        | 01111010 | 3.0        | 01111011 | 1.0        |
| 01111110 | 2.0        | 01111111 | 1.0        | 10100010 | 2.0        | 10100011 | 2.0        |
| 10100110 | 2.0        | 10100111 | 2.0        | 10101010 | 1.0        | 10101011 | 1.0        |
| 10101110 | 1.0        | 10101111 | 1.0        | 10110010 | 3.0        | 10110011 | 2.0        |
| 10110110 | 3.0        | 10110111 | 2.0        | 10111010 | 2.0        | 10111011 | 1.0        |
| 10111110 | 2.0        | 10111111 | 1.0        | 11000000 | 1.0        | 11000001 | 1.0        |
| 11000010 | 1.0        | 11000011 | 1.0        | 11010000 | 1.0        | 11010010 | 1.0        |
| 11011001 | -1.0       | 11011011 | -1.0       | 11100000 | 1.0        | 11100001 | 1.0        |
| 11100010 | 3.0        | 11100011 | 3.0        | 11100110 | 2.0        | 11100111 | 2.0        |
| 11101010 | 1.0        | 11101011 | 1.0        | 11101110 | 1.0        | 11101111 | 1.0        |
| 11110000 | 1.0        | 11110010 | 4.0        | 11110011 | 2.0        | 11110110 | 3.0        |
| 11110111 | 2.0        | 11111001 | -1.0       | 11111010 | 2.0        | 11111110 | 2.0        |
| 11111111 | 1.0        | Others   | 0.0        |          |            |          |            |

Table 36: Volume, Length: 93, Numbers: Number 0.0: 162, Number 1.0: 58, Number 2.0: 24, Number 3.0: 7, Number -1.0: 3, Number 4.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001100 | 1.0        | 00001101 | 1.0        | 00001110 | 2.0        | 00001111 | 1.0        |
| 00011100 | 1.0        | 00011101 | 1.0        | 00011110 | 2.0        | 00011111 | 1.0        |
| 00100000 | 1.0        | 00100001 | 1.0        | 00100010 | 1.0        | 00100100 | 1.0        |
| 00100101 | 1.0        | 00100110 | 1.0        | 00101000 | 1.0        | 00101001 | 1.0        |
| 00101010 | 1.0        | 00101100 | 2.0        | 00101101 | 2.0        | 00101110 | 3.0        |
| 00101111 | 1.0        | 00111100 | 1.0        | 00111101 | 1.0        | 00111110 | 2.0        |
| 00111111 | 1.0        | 01001100 | 1.0        | 01001101 | 1.0        | 01001110 | 2.0        |
| 01001111 | 1.0        | 01011100 | 1.0        | 01011101 | 1.0        | 01011110 | 2.0        |
| 01011111 | 1.0        | 01100000 | 1.0        | 01100001 | 1.0        | 01100010 | 1.0        |
| 01100100 | 1.0        | 01100101 | 1.0        | 01100110 | 1.0        | 01101000 | 1.0        |
| 01101001 | 1.0        | 01101010 | 1.0        | 01101100 | 2.0        | 01101101 | 2.0        |
| 01101110 | 3.0        | 01101111 | 1.0        | 01111100 | 1.0        | 01111101 | 1.0        |
| 01111110 | 2.0        | 01111111 | 1.0        | 10001100 | 2.0        | 10001101 | 2.0        |
| 10001110 | 3.0        | 10001111 | 2.0        | 10011100 | 2.0        | 10011101 | 2.0        |
| 10011110 | 3.0        | 10011111 | 2.0        | 10100000 | 1.0        | 10100001 | 1.0        |
| 10100010 | 1.0        | 10100100 | 1.0        | 10100101 | 1.0        | 10100110 | 1.0        |
| 10101000 | 1.0        | 10101001 | 1.0        | 10101010 | 1.0        | 10101100 | 3.0        |
| 10101101 | 3.0        | 10101110 | 4.0        | 10101111 | 2.0        | 10111100 | 2.0        |
| 10111101 | 2.0        | 10111110 | 3.0        | 10111111 | 2.0        | 11001100 | 1.0        |
| 11001101 | 1.0        | 11001110 | 2.0        | 11001111 | 1.0        | 11011100 | 1.0        |
| 11011101 | 1.0        | 11011110 | 2.0        | 11011111 | 1.0        | 11100011 | -1.0       |
| 11100111 | -1.0       | 11101011 | -1.0       | 11101100 | 1.0        | 11101101 | 1.0        |
| 11101110 | 2.0        | 11111100 | 1.0        | 11111101 | 1.0        | 11111110 | 2.0        |
| 11111111 | 1.0        | Others   | 0.0        |          |            |          |            |

### 1.11 Tables from Volume using OMP Regression model with 94 coefficients

Table 37: Volume, Length: 94, Numbers: Number 0.0: 161, Number 1.0: 65, Number 2.0: 24, Number 3.0: 5

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001111 | 1.0        | 00011111 | 1.0        | 00100000 | 1.0        | 00100001 | 1.0        |
| 00100100 | 1.0        | 00100101 | 1.0        | 00101000 | 1.0        | 00101001 | 1.0        |
| 00101100 | 1.0        | 00101101 | 1.0        | 00101111 | 1.0        | 00110000 | 1.0        |
| 00110001 | 2.0        | 00110011 | 1.0        | 00110100 | 1.0        | 00110101 | 2.0        |
| 00110111 | 1.0        | 00111000 | 1.0        | 00111001 | 2.0        | 00111011 | 1.0        |
| 00111100 | 1.0        | 00111101 | 2.0        | 00111111 | 2.0        | 01000100 | 1.0        |
| 01000101 | 1.0        | 01000110 | 1.0        | 01000111 | 1.0        | 01001100 | 1.0        |
| 01001101 | 1.0        | 01001110 | 1.0        | 01001111 | 2.0        | 01010000 | 1.0        |
| 01010010 | 1.0        | 01010100 | 2.0        | 01010101 | 1.0        | 01010110 | 2.0        |
| 01010111 | 1.0        | 01011000 | 1.0        | 01011010 | 1.0        | 01011100 | 2.0        |
| 01011101 | 1.0        | 01011110 | 2.0        | 01011111 | 2.0        | 01100000 | 1.0        |
| 01100001 | 1.0        | 01100100 | 2.0        | 01100101 | 2.0        | 01100110 | 1.0        |
| 01100111 | 1.0        | 01101000 | 1.0        | 01101001 | 1.0        | 01101100 | 2.0        |
| 01101101 | 2.0        | 01101110 | 1.0        | 01101111 | 2.0        | 01110000 | 2.0        |
| 01110001 | 2.0        | 01110010 | 1.0        | 01110011 | 1.0        | 01110100 | 3.0        |
| 01110101 | 3.0        | 01110110 | 2.0        | 01110111 | 2.0        | 01111000 | 2.0        |
| 01111001 | 2.0        | 01111010 | 1.0        | 01111011 | 1.0        | 01111100 | 3.0        |
| 01111101 | 3.0        | 01111110 | 2.0        | 01111111 | 3.0        | 10001111 | 1.0        |
| 10011111 | 1.0        | 10101111 | 1.0        | 10110001 | 1.0        | 10110011 | 1.0        |
| 10110101 | 1.0        | 10110111 | 1.0        | 10111001 | 1.0        | 10111011 | 1.0        |
| 10111101 | 1.0        | 10111111 | 2.0        | 11001111 | 1.0        | 11010000 | 1.0        |
| 11010010 | 1.0        | 11010100 | 1.0        | 11010110 | 1.0        | 11011000 | 1.0        |
| 11011010 | 1.0        | 11011100 | 1.0        | 11011110 | 1.0        | 11011111 | 1.0        |
| 11101111 | 1.0        | 11111111 | 1.0        | Others   | 0.0        |          |            |

Table 38: Volume, Length: 94, Numbers: Number 0.0: 161, Number 1.0: 64, Number 2.0: 26, Number 3.0: 4

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001000 | 1.0        | 00001001 | 1.0        | 00001010 | 1.0        | 00001011 | 2.0        |
| 00011000 | 1.0        | 00011001 | 1.0        | 00011010 | 1.0        | 00011011 | 2.0        |
| 00101000 | 1.0        | 00101001 | 1.0        | 00101010 | 1.0        | 00101011 | 2.0        |
| 00110000 | 1.0        | 00110001 | 1.0        | 00110010 | 2.0        | 00110011 | 1.0        |
| 00110100 | 1.0        | 00110101 | 1.0        | 00110110 | 2.0        | 00110111 | 1.0        |
| 00111000 | 2.0        | 00111001 | 2.0        | 00111010 | 3.0        | 00111011 | 3.0        |
| 00111100 | 1.0        | 00111101 | 1.0        | 00111110 | 2.0        | 00111111 | 1.0        |
| 01001000 | 1.0        | 01001001 | 1.0        | 01001010 | 1.0        | 01001011 | 2.0        |
| 01011000 | 1.0        | 01011001 | 1.0        | 01011010 | 1.0        | 01011011 | 2.0        |
| 01101000 | 1.0        | 01101001 | 1.0        | 01101010 | 1.0        | 01101011 | 2.0        |
| 01110000 | 1.0        | 01110001 | 1.0        | 01110010 | 2.0        | 01110011 | 1.0        |
| 01110100 | 1.0        | 01110101 | 1.0        | 01110110 | 2.0        | 01110111 | 1.0        |
| 01111000 | 2.0        | 01111001 | 2.0        | 01111010 | 3.0        | 01111011 | 3.0        |
| 01111100 | 1.0        | 01111101 | 1.0        | 01111110 | 2.0        | 01111111 | 1.0        |
| 10001000 | 1.0        | 10001001 | 1.0        | 10001010 | 1.0        | 10001011 | 2.0        |
| 10011000 | 1.0        | 10011001 | 1.0        | 10011010 | 1.0        | 10011011 | 2.0        |
| 10101000 | 1.0        | 10101001 | 1.0        | 10101010 | 1.0        | 10101011 | 2.0        |
| 10110010 | 1.0        | 10110110 | 1.0        | 10111000 | 1.0        | 10111001 | 1.0        |
| 10111010 | 2.0        | 10111011 | 2.0        | 10111110 | 1.0        | 11001011 | 1.0        |
| 11011011 | 1.0        | 11101011 | 1.0        | 11110000 | 1.0        | 11110001 | 1.0        |
| 11110010 | 2.0        | 11110011 | 1.0        | 11110100 | 1.0        | 11110101 | 1.0        |
| 11110110 | 2.0        | 11110111 | 1.0        | 11111000 | 1.0        | 11111001 | 1.0        |
| 11111010 | 2.0        | 11111011 | 2.0        | 11111100 | 1.0        | 11111101 | 1.0        |
| 11111110 | 2.0        | 11111111 | 1.0        | Others   | 0.0        |          |            |

## 1.12 Tables from Points using OMP Regression model with 98 coefficients

Table 39: Points, Length: 98, Numbers: Number 0.0: 157, Number 4.0: 47, Number 2.0: 22, Number 8.0: 13, Number -2.0: 6, Number -4.0: 4, Number 6.0: 3, Number 12.0: 2, Number 1.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001000 | 4.0        | 00001001 | 4.0        | 00001010 | 4.0        | 00001011 | 4.0        |
| 00010000 | 4.0        | 00010010 | 4.0        | 00011000 | 8.0        | 00011001 | 4.0        |
| 00011010 | 8.0        | 00011011 | 4.0        | 00100010 | 4.0        | 00100110 | 4.0        |
| 00101000 | 8.0        | 00101001 | 8.0        | 00101010 | 8.0        | 00101011 | 4.0        |
| 00101100 | 4.0        | 00101101 | 4.0        | 00101110 | 4.0        | 00110000 | 4.0        |
| 00110010 | 8.0        | 00110011 | 2.0        | 00110110 | 4.0        | 00110111 | 2.0        |
| 00111000 | 12.0       | 00111001 | 8.0        | 00111010 | 12.0       | 00111011 | 6.0        |
| 00111100 | 4.0        | 00111101 | 4.0        | 00111110 | 4.0        | 00111111 | 2.0        |
| 01001000 | 4.0        | 01001001 | 4.0        | 01001010 | 4.0        | 01001011 | 4.0        |
| 01011000 | 4.0        | 01011001 | 4.0        | 01011010 | 4.0        | 01011011 | 4.0        |
| 01100010 | 4.0        | 01100110 | 4.0        | 01101000 | 8.0        | 01101001 | 8.0        |
| 01101010 | 8.0        | 01101011 | 4.0        | 01101100 | 4.0        | 01101101 | 4.0        |
| 01101110 | 4.0        | 01110010 | 4.0        | 01110011 | 2.0        | 01110110 | 4.0        |
| 01110111 | 2.0        | 01111000 | 8.0        | 01111001 | 8.0        | 01111010 | 8.0        |
| 01111011 | 6.0        | 01111100 | 4.0        | 01111101 | 4.0        | 01111110 | 4.0        |
| 01111111 | 2.0        | 10010000 | 4.0        | 10010010 | 4.0        | 10011000 | 4.0        |
| 10011010 | 4.0        | 10100011 | -4.0       | 10100111 | -4.0       | 10101010 | 2.0        |
| 10101011 | -2.0       | 10101110 | 2.0        | 10101111 | -2.0       | 10110000 | 4.0        |
| 10110010 | 4.0        | 10110011 | -2.0       | 10110111 | -2.0       | 10111000 | 4.0        |
| 10111010 | 6.0        | 10111110 | 2.0        | 11100011 | -4.0       | 11100111 | -4.0       |
| 11101010 | 2.0        | 11101011 | -2.0       | 11101110 | 2.0        | 11101111 | -2.0       |
| 11110000 | 2.0        | 11110001 | 2.0        | 11110010 | 2.0        | 11110100 | 2.0        |
| 11110101 | 2.0        | 11110110 | 2.0        | 11111000 | 2.0        | 11111001 | 2.0        |
| 11111010 | 4.0        | 11111011 | 2.0        | 11111100 | 2.0        | 11111101 | 2.0        |
| 11111110 | 4.0        | 11111111 | 1.0        | Others   | 0.0        |          |            |

Table 40: Points, Length: 98, Numbers: Number 0.0: 157, Number 4.0: 47, Number 2.0: 22, Number 8.0: 13, Number -2.0: 6, Number -4.0: 4, Number 6.0: 3, Number 12.0: 2, Number 1.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 4.0        | 00000110 | 4.0        | 00001010 | 4.0        | 00001110 | 4.0        |
| 00001111 | 2.0        | 00011111 | 2.0        | 00101111 | 2.0        | 00111111 | 2.0        |
| 01000000 | 4.0        | 01000001 | 4.0        | 01000010 | 8.0        | 01000011 | 4.0        |
| 01000110 | 4.0        | 01001000 | 8.0        | 01001001 | 8.0        | 01001010 | 12.0       |
| 01001011 | 8.0        | 01001110 | 4.0        | 01001111 | 2.0        | 01011000 | 4.0        |
| 01011001 | 4.0        | 01011010 | 4.0        | 01011011 | 4.0        | 01011111 | 2.0        |
| 01100000 | 4.0        | 01100001 | 4.0        | 01100010 | 4.0        | 01100011 | 4.0        |
| 01101000 | 8.0        | 01101001 | 8.0        | 01101010 | 8.0        | 01101011 | 8.0        |
| 01101111 | 2.0        | 01111000 | 4.0        | 01111001 | 4.0        | 01111010 | 4.0        |
| 01111011 | 4.0        | 01111111 | 2.0        | 10000010 | 4.0        | 10000110 | 4.0        |
| 10001000 | 4.0        | 10001001 | 4.0        | 10001010 | 8.0        | 10001011 | 4.0        |
| 10001110 | 4.0        | 10001111 | 2.0        | 10011000 | 4.0        | 10011001 | 4.0        |
| 10011010 | 4.0        | 10011011 | 4.0        | 10011111 | 2.0        | 10101010 | 2.0        |
| 10101011 | 2.0        | 10101100 | -4.0       | 10101101 | -4.0       | 10101110 | -2.0       |
| 10111010 | 2.0        | 10111011 | 2.0        | 10111100 | -4.0       | 10111101 | -4.0       |
| 10111110 | -2.0       | 11000000 | 4.0        | 11000001 | 4.0        | 11000010 | 8.0        |
| 11000011 | 4.0        | 11000110 | 4.0        | 11001000 | 8.0        | 11001001 | 8.0        |
| 11001010 | 12.0       | 11001011 | 8.0        | 11001100 | 2.0        | 11001101 | 2.0        |
| 11001110 | 6.0        | 11001111 | 4.0        | 11011000 | 4.0        | 11011001 | 4.0        |
| 11011010 | 4.0        | 11011011 | 4.0        | 11011100 | 2.0        | 11011101 | 2.0        |
| 11011110 | 2.0        | 11011111 | 4.0        | 11100000 | 4.0        | 11100001 | 4.0        |
| 11100010 | 4.0        | 11100011 | 4.0        | 11101000 | 4.0        | 11101001 | 4.0        |
| 11101010 | 6.0        | 11101011 | 6.0        | 11101100 | -2.0       | 11101101 | -2.0       |
| 11101111 | 2.0        | 11111010 | 2.0        | 11111011 | 2.0        | 11111100 | -2.0       |
| 11111101 | -2.0       | 11111111 | 1.0        | Others   | 0.0        |          |            |

### 1.13 Tables from Points using OMP Regression model with 109 coefficients

Table 41: Points, Length: 109, Numbers: Number 0.0: 146, Number 4.0: 49, Number -2.0: 24, Number 8.0: 13, Number 2.0: 13, Number 6.0: 6, Number 12.0: 3, Number 1.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 4.0        | 00000110 | 4.0        | 00001010 | 4.0        | 00001110 | 4.0        |
| 00010010 | 4.0        | 00010110 | 4.0        | 00011010 | 4.0        | 00011110 | 4.0        |
| 00100010 | 4.0        | 00100110 | 4.0        | 00101010 | 4.0        | 00101110 | 4.0        |
| 01000000 | 4.0        | 01000010 | 8.0        | 01000110 | 4.0        | 01001000 | 4.0        |
| 01001010 | 8.0        | 01001110 | 4.0        | 01010010 | 4.0        | 01010100 | -2.0       |
| 01010101 | -2.0       | 01010110 | 2.0        | 01010111 | -2.0       | 01011010 | 4.0        |
| 01011100 | -2.0       | 01011101 | -2.0       | 01011110 | 2.0        | 01011111 | -2.0       |
| 01100000 | 4.0        | 01100010 | 8.0        | 01100110 | 4.0        | 01101000 | 4.0        |
| 01101010 | 8.0        | 01101110 | 4.0        | 01110100 | -2.0       | 01110101 | -2.0       |
| 01110110 | -2.0       | 01110111 | -2.0       | 01111100 | -2.0       | 01111101 | -2.0       |
| 01111110 | -2.0       | 01111111 | -2.0       | 10000010 | 8.0        | 10000011 | 4.0        |
| 10000110 | 8.0        | 10000111 | 4.0        | 10001010 | 4.0        | 10001110 | 4.0        |
| 10010010 | 8.0        | 10010011 | 4.0        | 10010110 | 8.0        | 10010111 | 4.0        |
| 10011010 | 4.0        | 10011110 | 4.0        | 10100010 | 4.0        | 10100110 | 4.0        |
| 10101000 | -2.0       | 10101001 | -2.0       | 10101010 | 4.0        | 10101100 | -2.0       |
| 10101101 | -2.0       | 10101110 | 4.0        | 10111000 | -2.0       | 10111001 | -2.0       |
| 10111100 | -2.0       | 10111101 | -2.0       | 11000000 | 4.0        | 11000010 | 12.0       |
| 11000011 | 4.0        | 11000110 | 8.0        | 11000111 | 4.0        | 11001000 | 8.0        |
| 11001001 | 4.0        | 11001010 | 12.0       | 11001011 | 4.0        | 11001100 | 2.0        |
| 11001101 | 2.0        | 11001110 | 6.0        | 11001111 | 2.0        | 11010010 | 8.0        |
| 11010011 | 4.0        | 11010100 | -2.0       | 11010101 | -2.0       | 11010110 | 6.0        |
| 11010111 | 2.0        | 11011000 | 4.0        | 11011001 | 4.0        | 11011010 | 8.0        |
| 11011011 | 4.0        | 11011110 | 4.0        | 11100000 | 4.0        | 11100010 | 8.0        |
| 11100110 | 4.0        | 11101000 | 6.0        | 11101001 | 2.0        | 11101010 | 12.0       |
| 11101011 | 4.0        | 11101110 | 6.0        | 11101111 | 2.0        | 11110000 | 2.0        |
| 11110001 | 2.0        | 11110010 | 2.0        | 11110011 | 2.0        | 11111000 | 4.0        |
| 11111001 | 4.0        | 11111010 | 6.0        | 11111011 | 6.0        | 11111110 | 2.0        |
| 11111111 | 1.0        | Others   | 0.0        |          |            |          |            |



Table 42: Points, Length: 109, Numbers: Number 0.0: 146, Number 2.0: 46, Number 4.0: 28, Number 6.0: 19, Number -2.0: 9, Number 8.0: 5, Number 10.0: 1, Number 1.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 6.0        | 00000011 | 4.0        | 00000100 | 2.0        | 00000110 | 8.0        |
| 00000111 | 6.0        | 00001010 | 2.0        | 00001110 | 2.0        | 00001111 | 2.0        |
| 00010001 | 4.0        | 00010010 | 6.0        | 00010011 | 6.0        | 00010100 | 2.0        |
| 00010110 | 8.0        | 00010111 | 4.0        | 00011001 | 4.0        | 00011010 | 2.0        |
| 00011011 | 2.0        | 00011110 | 2.0        | 00100011 | -2.0       | 00100100 | 2.0        |
| 00100110 | 2.0        | 00101011 | -2.0       | 00110001 | 4.0        | 00110011 | 2.0        |
| 00110100 | 2.0        | 00110110 | 2.0        | 00111001 | 4.0        | 00111011 | 2.0        |
| 01000001 | 6.0        | 01000010 | 6.0        | 01000011 | 10.0       | 01000110 | 6.0        |
| 01000111 | 6.0        | 01001001 | 6.0        | 01001010 | 2.0        | 01001011 | 6.0        |
| 01001101 | 2.0        | 01001110 | 2.0        | 01001111 | 4.0        | 01010000 | 2.0        |
| 01010001 | 6.0        | 01010010 | 8.0        | 01010011 | 8.0        | 01010100 | 2.0        |
| 01010101 | 2.0        | 01010110 | 8.0        | 01010111 | 6.0        | 01011000 | 2.0        |
| 01011001 | 6.0        | 01011010 | 4.0        | 01011011 | 4.0        | 01011100 | 2.0        |
| 01011101 | 4.0        | 01011110 | 4.0        | 01011111 | 4.0        | 01100001 | 6.0        |
| 01100011 | 4.0        | 01101001 | 6.0        | 01101011 | 4.0        | 01101101 | 2.0        |
| 01101111 | 2.0        | 01110001 | 4.0        | 01110011 | 2.0        | 01111001 | 4.0        |
| 01111011 | 2.0        | 01111101 | 2.0        | 01111111 | 2.0        | 10000011 | -2.0       |
| 10000100 | 2.0        | 10000110 | 2.0        | 10001011 | -2.0       | 10010001 | 4.0        |
| 10010100 | 2.0        | 10010110 | 2.0        | 10010111 | -2.0       | 10011001 | 4.0        |
| 10011111 | -2.0       | 10100011 | -2.0       | 10100100 | 2.0        | 10100110 | 2.0        |
| 10101011 | -2.0       | 10110001 | 4.0        | 10110011 | 2.0        | 10110100 | 2.0        |
| 10110110 | 2.0        | 10111001 | 4.0        | 10111011 | 2.0        | 11000001 | 6.0        |
| 11000011 | 4.0        | 11001001 | 6.0        | 11001011 | 4.0        | 11001101 | 2.0        |
| 11001111 | 2.0        | 11010001 | 4.0        | 11010111 | -2.0       | 11011001 | 4.0        |
| 11011101 | 2.0        | 11100001 | 6.0        | 11100011 | 4.0        | 11101001 | 6.0        |
| 11101011 | 4.0        | 11101101 | 2.0        | 11101111 | 2.0        | 11110001 | 4.0        |
| 11110011 | 2.0        | 11111001 | 4.0        | 11111011 | 2.0        | 11111101 | 2.0        |
| 11111111 | 1.0        | Others   | 0.0        |          |            |          |            |

### 1.14 Tables from Edges using OMP Regression model with 113 coefficients

Table 43: Edges, Length: 113, Numbers: Number 0.0: 142, Number 8.0: 22, Number 12.0: 19, Number 4.0: 15, Number 5.0: 13, Number -3.0: 6, Number -4.0: 6, Number 16.0: 4, Number 13.0: 4, Number 20.0: 4, Number 10.0: 4, Number 1.0: 4, Number 24.0: 3, Number -8.0: 1, Number 17.0: 1, Number 29.0: 1, Number -11.0: 1, Number -12.0: 1, Number 2.0: 1, Number -9.0: 1, Number 6.0: 1, Number 3.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | 8.0        | 00000011 | 8.0        | 00001001 | 8.0        | 00010001 | 8.0        |
| 00010011 | 8.0        | 00011001 | 8.0        | 00100001 | 12.0       | 00100011 | 8.0        |
| 00100101 | 4.0        | 00101001 | 12.0       | 00101101 | 4.0        | 00110001 | 8.0        |
| 00110011 | 5.0        | 00110111 | -3.0       | 00111001 | 8.0        | 00111011 | -3.0       |
| 00111111 | -3.0       | 01000001 | 8.0        | 01000011 | 8.0        | 01001001 | 8.0        |
| 01011011 | -8.0       | 01100001 | 12.0       | 01100011 | 8.0        | 01100101 | 4.0        |
| 01101001 | 12.0       | 01101101 | 4.0        | 01110011 | -3.0       | 01110111 | -3.0       |
| 01111011 | -11.0      | 01111111 | -3.0       | 10000000 | 4.0        | 10000001 | 12.0       |
| 10000010 | 4.0        | 10000011 | 12.0       | 10001001 | 8.0        | 10010000 | 4.0        |
| 10010001 | 12.0       | 10010010 | 4.0        | 10010011 | 12.0       | 10011001 | 8.0        |
| 10100000 | 8.0        | 10100001 | 20.0       | 10100010 | 16.0       | 10100011 | 24.0       |
| 10100100 | 4.0        | 10100101 | 8.0        | 10100110 | 12.0       | 10100111 | 12.0       |
| 10101000 | 4.0        | 10101001 | 16.0       | 10101010 | 5.0        | 10101011 | 5.0        |
| 10101100 | 4.0        | 10101101 | 8.0        | 10101110 | 5.0        | 10101111 | 5.0        |
| 10110000 | 16.0       | 10110001 | 24.0       | 10110010 | 24.0       | 10110011 | 29.0       |
| 10110100 | 12.0       | 10110101 | 12.0       | 10110110 | 20.0       | 10110111 | 17.0       |
| 10111000 | 12.0       | 10111001 | 20.0       | 10111010 | 13.0       | 10111011 | 10.0       |
| 10111100 | 12.0       | 10111101 | 12.0       | 10111110 | 13.0       | 10111111 | 10.0       |
| 11000001 | 8.0        | 11000011 | 8.0        | 11001000 | -4.0       | 11001001 | 4.0        |
| 11001010 | -4.0       | 11001011 | -4.0       | 11011000 | -4.0       | 11011001 | -4.0       |
| 11011010 | -4.0       | 11011011 | -12.0      | 11100000 | 4.0        | 11100001 | 16.0       |
| 11100010 | 12.0       | 11100011 | 20.0       | 11100100 | 4.0        | 11100101 | 8.0        |
| 11100110 | 12.0       | 11100111 | 12.0       | 11101001 | 12.0       | 11101010 | 1.0        |
| 11101011 | 1.0        | 11101100 | 4.0        | 11101101 | 8.0        | 11101110 | 5.0        |
| 11101111 | 5.0        | 11110000 | 5.0        | 11110001 | 5.0        | 11110010 | 13.0       |
| 11110011 | 10.0       | 11110100 | 5.0        | 11110101 | 5.0        | 11110110 | 13.0       |
| 11110111 | 10.0       | 11111000 | 1.0        | 11111001 | 1.0        | 11111010 | 2.0        |
| 11111011 | -9.0       | 11111100 | 5.0        | 11111101 | 5.0        | 11111110 | 6.0        |
| 11111111 | 3.0        | Others   | 0.0        |          |            |          |            |

Table 44: Edges, Length: 113, Numbers: Number 0.0: 142, Number 8.0: 22, Number 12.0: 19, Number 4.0: 15, Number 5.0: 13, Number -4.0: 6, Number -3.0: 6, Number 16.0: 4, Number 20.0: 4, Number 1.0: 4, Number 10.0: 4, Number 13.0: 4, Number 24.0: 3, Number 2.0: 1, Number -8.0: 1, Number 6.0: 1, Number -12.0: 1, Number 29.0: 1, Number 17.0: 1, Number -11.0: 1, Number -9.0: 1, Number 3.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 4.0        | 00001011 | -4.0       | 00010010 | 4.0        | 00011011 | -4.0       |
| 00100010 | 8.0        | 00100011 | 4.0        | 00100110 | 4.0        | 00100111 | 4.0        |
| 00101010 | 4.0        | 00101110 | 4.0        | 00101111 | 4.0        | 00110010 | 16.0       |
| 00110011 | 5.0        | 00110110 | 12.0       | 00110111 | 5.0        | 00111010 | 12.0       |
| 00111011 | 1.0        | 00111110 | 12.0       | 00111111 | 5.0        | 01000000 | 8.0        |
| 01000001 | 8.0        | 01000010 | 12.0       | 01000011 | 8.0        | 01001000 | 8.0        |
| 01001001 | 8.0        | 01001010 | 8.0        | 01001011 | 4.0        | 01010000 | 8.0        |
| 01010010 | 12.0       | 01011000 | 8.0        | 01011010 | 8.0        | 01011011 | -4.0       |
| 01100000 | 12.0       | 01100001 | 12.0       | 01100010 | 20.0       | 01100011 | 16.0       |
| 01100100 | 4.0        | 01100101 | 4.0        | 01100110 | 8.0        | 01100111 | 8.0        |
| 01101000 | 12.0       | 01101001 | 12.0       | 01101010 | 16.0       | 01101011 | 12.0       |
| 01101100 | 4.0        | 01101101 | 4.0        | 01101110 | 8.0        | 01101111 | 8.0        |
| 01110000 | 8.0        | 01110010 | 24.0       | 01110011 | 5.0        | 01110110 | 12.0       |
| 01110111 | 5.0        | 01111000 | 8.0        | 01111010 | 20.0       | 01111011 | 1.0        |
| 01111110 | 12.0       | 01111111 | 5.0        | 10000010 | 4.0        | 10001011 | -4.0       |
| 10010010 | 4.0        | 10011011 | -4.0       | 10100010 | 16.0       | 10100011 | 12.0       |
| 10100110 | 12.0       | 10100111 | 12.0       | 10101010 | 5.0        | 10101011 | 1.0        |
| 10101110 | 5.0        | 10101111 | 5.0        | 10110010 | 24.0       | 10110011 | 13.0       |
| 10110110 | 20.0       | 10110111 | 13.0       | 10111010 | 13.0       | 10111011 | 2.0        |
| 10111110 | 13.0       | 10111111 | 6.0        | 11000000 | 8.0        | 11000001 | 8.0        |
| 11000010 | 12.0       | 11000011 | 8.0        | 11001011 | -4.0       | 11010000 | 8.0        |
| 11010010 | 12.0       | 11011001 | -8.0       | 11011011 | -12.0      | 11100000 | 8.0        |
| 11100001 | 8.0        | 11100010 | 24.0       | 11100011 | 20.0       | 11100110 | 12.0       |
| 11100111 | 12.0       | 11101010 | 5.0        | 11101011 | 1.0        | 11101110 | 5.0        |
| 11101111 | 5.0        | 11110000 | 5.0        | 11110001 | -3.0       | 11110010 | 29.0       |
| 11110011 | 10.0       | 11110100 | -3.0       | 11110101 | -3.0       | 11110110 | 17.0       |
| 11110111 | 10.0       | 11111000 | -3.0       | 11111001 | -11.0      | 11111010 | 10.0       |
| 11111011 | -9.0       | 11111100 | -3.0       | 11111101 | -3.0       | 11111110 | 10.0       |
| 11111111 | 3.0        | Others   | 0.0        |          |            |          |            |

### 1.15 Tables from Edges using OMP Regression model with 114 coefficients

Table 45: Edges, Length: 114, Numbers: Number 0.0: 141, Number 8.0: 25, Number 12.0: 23, Number 4.0: 17, Number 5.0: 13, Number 1.0: 6, Number -3.0: 5, Number -4.0: 5, Number 16.0: 3, Number 13.0: 3, Number 10.0: 3, Number 3.0: 2, Number 17.0: 2, Number 20.0: 2, Number 9.0: 1, Number 24.0: 1, Number -2.0: 1, Number 2.0: 1, Number 6.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | 8.0        | 00000011 | 8.0        | 00000101 | 8.0        | 00000111 | 8.0        |
| 00001001 | 12.0       | 00001011 | 8.0        | 00001101 | 8.0        | 00001111 | 5.0        |
| 00011001 | 4.0        | 00011111 | -3.0       | 00100001 | 8.0        | 00100101 | 8.0        |
| 00101001 | 12.0       | 00101101 | 8.0        | 00101111 | -3.0       | 00111001 | 4.0        |
| 00111111 | -3.0       | 01000001 | 8.0        | 01000011 | 8.0        | 01000101 | 12.0       |
| 01000111 | 12.0       | 01001001 | 12.0       | 01001011 | 8.0        | 01001101 | 12.0       |
| 01001111 | 9.0        | 01011001 | 4.0        | 01011111 | -3.0       | 01100001 | 8.0        |
| 01100101 | 12.0       | 01100111 | 4.0        | 01101001 | 12.0       | 01101101 | 12.0       |
| 01101111 | 1.0        | 01111001 | 4.0        | 01111111 | -3.0       | 10000000 | 4.0        |
| 10000001 | 12.0       | 10000010 | 4.0        | 10000011 | 12.0       | 10000100 | 4.0        |
| 10000101 | 12.0       | 10000110 | 4.0        | 10000111 | 12.0       | 10001000 | 8.0        |
| 10001001 | 20.0       | 10001010 | 4.0        | 10001011 | 12.0       | 10001100 | 16.0       |
| 10001101 | 24.0       | 10001110 | 12.0       | 10001111 | 17.0       | 10011000 | 4.0        |
| 10011001 | 8.0        | 10011100 | 12.0       | 10011101 | 12.0       | 10011110 | 8.0        |
| 10011111 | 5.0        | 10100001 | 8.0        | 10100101 | 8.0        | 10101000 | 4.0        |
| 10101001 | 16.0       | 10101010 | 5.0        | 10101011 | 5.0        | 10101100 | 12.0       |
| 10101101 | 20.0       | 10101110 | 13.0       | 10101111 | 10.0       | 10111000 | 4.0        |
| 10111001 | 8.0        | 10111010 | 5.0        | 10111011 | 5.0        | 10111100 | 12.0       |
| 10111101 | 12.0       | 10111110 | 13.0       | 10111111 | 10.0       | 11000001 | 8.0        |
| 11000011 | 8.0        | 11000101 | 12.0       | 11000111 | 12.0       | 11001000 | 4.0        |
| 11001001 | 16.0       | 11001011 | 8.0        | 11001100 | 5.0        | 11001101 | 17.0       |
| 11001110 | 1.0        | 11001111 | 10.0       | 11011000 | 4.0        | 11011001 | 8.0        |
| 11011100 | 5.0        | 11011101 | 5.0        | 11011110 | 1.0        | 11011111 | -2.0       |
| 11100000 | -4.0       | 11100001 | 4.0        | 11100010 | -4.0       | 11100011 | -4.0       |
| 11100100 | -4.0       | 11100101 | 8.0        | 11100110 | -4.0       | 11101001 | 12.0       |
| 11101010 | 1.0        | 11101011 | 1.0        | 11101100 | 1.0        | 11101101 | 13.0       |
| 11101110 | 2.0        | 11101111 | 3.0        | 11111000 | 4.0        | 11111001 | 8.0        |
| 11111010 | 5.0        | 11111011 | 5.0        | 11111100 | 5.0        | 11111101 | 5.0        |
| 11111110 | 6.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

Table 46: Edges, Length: 114, Numbers: Number 0.0: 141, Number 8.0: 25, Number 12.0: 23, Number 4.0: 17, Number 5.0: 13, Number 1.0: 6, Number -3.0: 5, Number -4.0: 5, Number 16.0: 3, Number 10.0: 3, Number 13.0: 3, Number 3.0: 2, Number 17.0: 2, Number 20.0: 2, Number 9.0: 1, Number 24.0: 1, Number -2.0: 1, Number 2.0: 1, Number 6.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000001 | 8.0        | 00000011 | 8.0        | 00001001 | 8.0        | 00001011 | 12.0       |
| 00010001 | 8.0        | 00010011 | 8.0        | 00011001 | 8.0        | 00011011 | 12.0       |
| 00100001 | 12.0       | 00100011 | 8.0        | 00100101 | 4.0        | 00101001 | 12.0       |
| 00101011 | 12.0       | 00101101 | 4.0        | 00110001 | 8.0        | 00110011 | 5.0        |
| 00110111 | -3.0       | 00111001 | 8.0        | 00111011 | 9.0        | 00111111 | -3.0       |
| 01000001 | 8.0        | 01000011 | 8.0        | 01001001 | 8.0        | 01001011 | 12.0       |
| 01011011 | 4.0        | 01100001 | 12.0       | 01100011 | 8.0        | 01100101 | 4.0        |
| 01101001 | 12.0       | 01101011 | 12.0       | 01101101 | 4.0        | 01110011 | -3.0       |
| 01110111 | -3.0       | 01111011 | 1.0        | 01111111 | -3.0       | 10000000 | 4.0        |
| 10000001 | 12.0       | 10000010 | 4.0        | 10000011 | 12.0       | 10001001 | 8.0        |
| 10001011 | 12.0       | 10010000 | 4.0        | 10010001 | 12.0       | 10010010 | 4.0        |
| 10010011 | 12.0       | 10011001 | 8.0        | 10011011 | 12.0       | 10100000 | 8.0        |
| 10100001 | 20.0       | 10100010 | 16.0       | 10100011 | 24.0       | 10100100 | 4.0        |
| 10100101 | 8.0        | 10100110 | 12.0       | 10100111 | 12.0       | 10101000 | 4.0        |
| 10101001 | 16.0       | 10101010 | 5.0        | 10101011 | 17.0       | 10101100 | 4.0        |
| 10101101 | 8.0        | 10101110 | 5.0        | 10101111 | 5.0        | 10110000 | 4.0        |
| 10110001 | 12.0       | 10110010 | 12.0       | 10110011 | 17.0       | 10110110 | 8.0        |
| 10110111 | 5.0        | 10111001 | 8.0        | 10111010 | 1.0        | 10111011 | 10.0       |
| 10111110 | 1.0        | 10111111 | -2.0       | 11000001 | 8.0        | 11000011 | 8.0        |
| 11001000 | -4.0       | 11001001 | 4.0        | 11001010 | -4.0       | 11001011 | 8.0        |
| 11011000 | -4.0       | 11011001 | -4.0       | 11011010 | -4.0       | 11100000 | 4.0        |
| 11100001 | 16.0       | 11100010 | 12.0       | 11100011 | 20.0       | 11100100 | 4.0        |
| 11100101 | 8.0        | 11100110 | 12.0       | 11100111 | 12.0       | 11101001 | 12.0       |
| 11101010 | 1.0        | 11101011 | 13.0       | 11101100 | 4.0        | 11101101 | 8.0        |
| 11101110 | 5.0        | 11101111 | 5.0        | 11110000 | 5.0        | 11110001 | 5.0        |
| 11110010 | 13.0       | 11110011 | 10.0       | 11110100 | 5.0        | 11110101 | 5.0        |
| 11110110 | 13.0       | 11110111 | 10.0       | 11111000 | 1.0        | 11111001 | 1.0        |
| 11111010 | 2.0        | 11111011 | 3.0        | 11111100 | 5.0        | 11111101 | 5.0        |
| 11111110 | 6.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

## 1.16 Tables from Faces using OMP Regression model with 114 coefficients

Table 47: Faces, Length: 114, Numbers: Number 0.0: 141, Number 5.0: 24, Number 4.0: 22, Number 9.0: 15, Number 6.0: 11, Number -1.0: 11, Number 1.0: 9, Number 3.0: 8, Number 8.0: 5, Number 10.0: 4, Number 11.0: 1, Number 15.0: 1, Number 14.0: 1, Number 13.0: 1, Number 12.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000100 | 1.0        | 00000110 | 1.0        | 00001100 | 5.0        | 00001101 | 4.0        |
| 00001110 | 1.0        | 00001111 | 4.0        | 00011100 | 4.0        | 00011101 | 4.0        |
| 00011111 | 4.0        | 00100000 | 5.0        | 00100001 | 5.0        | 00100010 | 5.0        |
| 00100100 | 6.0        | 00100101 | 5.0        | 00100110 | 6.0        | 00101000 | 6.0        |
| 00101001 | 6.0        | 00101010 | 5.0        | 00101100 | 11.0       | 00101101 | 10.0       |
| 00101110 | 6.0        | 00101111 | 4.0        | 00111000 | 1.0        | 00111001 | 1.0        |
| 00111100 | 5.0        | 00111101 | 5.0        | 00111111 | 4.0        | 01000101 | -1.0       |
| 01000111 | -1.0       | 01001100 | 4.0        | 01001101 | 3.0        | 01001111 | 3.0        |
| 01011100 | 4.0        | 01011101 | 4.0        | 01011111 | 4.0        | 01100000 | 5.0        |
| 01100001 | 5.0        | 01100010 | 5.0        | 01100100 | 5.0        | 01100101 | 4.0        |
| 01100110 | 5.0        | 01100111 | -1.0       | 01101000 | 6.0        | 01101001 | 6.0        |
| 01101010 | 5.0        | 01101100 | 10.0       | 01101101 | 9.0        | 01101110 | 5.0        |
| 01101111 | 3.0        | 01111000 | 1.0        | 01111001 | 1.0        | 01111100 | 5.0        |
| 01111101 | 5.0        | 01111111 | 4.0        | 10000100 | 1.0        | 10000110 | 1.0        |
| 10001100 | 10.0       | 10001101 | 9.0        | 10001110 | 6.0        | 10001111 | 9.0        |
| 10011100 | 9.0        | 10011101 | 9.0        | 10011110 | 5.0        | 10011111 | 9.0        |
| 10100000 | 5.0        | 10100001 | 5.0        | 10100010 | 5.0        | 10100100 | 6.0        |
| 10100101 | 5.0        | 10100110 | 6.0        | 10101000 | 5.0        | 10101001 | 5.0        |
| 10101010 | 4.0        | 10101011 | -1.0       | 10101100 | 15.0       | 10101101 | 14.0       |
| 10101110 | 10.0       | 10101111 | 8.0        | 10111010 | -1.0       | 10111011 | -1.0       |
| 10111100 | 9.0        | 10111101 | 9.0        | 10111110 | 4.0        | 10111111 | 8.0        |
| 11000101 | -1.0       | 11000111 | -1.0       | 11001100 | 4.0        | 11001101 | 3.0        |
| 11001111 | 3.0        | 11011100 | 4.0        | 11011101 | 4.0        | 11011111 | 4.0        |
| 11100000 | 9.0        | 11100001 | 9.0        | 11100010 | 9.0        | 11100011 | 4.0        |
| 11100100 | 9.0        | 11100101 | 8.0        | 11100110 | 9.0        | 11100111 | 3.0        |
| 11101000 | 9.0        | 11101001 | 9.0        | 11101010 | 8.0        | 11101011 | 3.0        |
| 11101100 | 13.0       | 11101101 | 12.0       | 11101110 | 8.0        | 11101111 | 6.0        |
| 11111010 | -1.0       | 11111011 | -1.0       | 11111100 | 4.0        | 11111101 | 4.0        |
| 11111110 | -1.0       | 11111111 | 3.0        | Others   | 0.0        |          |            |

Table 48: Faces, Length: 114, Numbers: Number 0.0: 141, Number 5.0: 24, Number 4.0: 22, Number 9.0: 15, Number 6.0: 11, Number -1.0: 11, Number 1.0: 9, Number 3.0: 8, Number 8.0: 5, Number 10.0: 4, Number 11.0: 1, Number 13.0: 1, Number 15.0: 1, Number 14.0: 1, Number 12.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001000 | 5.0        | 00001001 | 6.0        | 00001010 | 5.0        | 00001011 | 5.0        |
| 00001100 | 5.0        | 00001101 | 5.0        | 00001110 | 5.0        | 00001111 | 4.0        |
| 00010000 | 1.0        | 00010001 | 5.0        | 00010010 | 1.0        | 00010011 | 10.0       |
| 00010100 | 1.0        | 00010101 | 1.0        | 00010110 | 1.0        | 00010111 | 6.0        |
| 00011000 | 6.0        | 00011001 | 11.0       | 00011010 | 6.0        | 00011011 | 15.0       |
| 00011100 | 6.0        | 00011101 | 6.0        | 00011110 | 6.0        | 00011111 | 10.0       |
| 00101000 | 5.0        | 00101001 | 6.0        | 00101010 | 9.0        | 00101011 | 9.0        |
| 00101100 | 5.0        | 00101101 | 5.0        | 00101110 | 9.0        | 00101111 | 8.0        |
| 00110001 | 4.0        | 00110011 | 4.0        | 00111000 | 5.0        | 00111001 | 10.0       |
| 00111010 | 9.0        | 00111011 | 13.0       | 00111100 | 5.0        | 00111101 | 5.0        |
| 00111110 | 9.0        | 00111111 | 8.0        | 01001000 | 5.0        | 01001001 | 6.0        |
| 01001010 | 5.0        | 01001011 | 5.0        | 01001111 | -1.0       | 01010001 | 4.0        |
| 01010011 | 9.0        | 01010101 | 4.0        | 01010111 | 9.0        | 01011000 | 5.0        |
| 01011001 | 10.0       | 01011010 | 5.0        | 01011011 | 14.0       | 01011101 | 4.0        |
| 01011111 | 8.0        | 01101000 | 5.0        | 01101001 | 6.0        | 01101010 | 9.0        |
| 01101011 | 9.0        | 01101110 | 4.0        | 01101111 | 3.0        | 01110000 | -1.0       |
| 01110001 | 3.0        | 01110010 | -1.0       | 01110011 | 3.0        | 01110100 | -1.0       |
| 01110101 | 3.0        | 01110110 | -1.0       | 01110111 | 3.0        | 01111000 | 4.0        |
| 01111001 | 9.0        | 01111010 | 8.0        | 01111011 | 12.0       | 01111100 | -1.0       |
| 01111101 | 3.0        | 01111110 | 3.0        | 01111111 | 6.0        | 10001001 | 1.0        |
| 10001111 | -1.0       | 10010001 | 4.0        | 10010011 | 9.0        | 10010111 | 5.0        |
| 10011001 | 5.0        | 10011011 | 9.0        | 10011111 | 4.0        | 10101001 | 1.0        |
| 10101111 | -1.0       | 10110001 | 4.0        | 10110011 | 4.0        | 10111001 | 5.0        |
| 10111011 | 4.0        | 10111111 | -1.0       | 11001001 | 1.0        | 11001111 | -1.0       |
| 11010001 | 4.0        | 11010011 | 9.0        | 11010101 | 4.0        | 11010111 | 9.0        |
| 11011001 | 5.0        | 11011011 | 9.0        | 11011101 | 4.0        | 11011111 | 8.0        |
| 11101001 | 1.0        | 11101111 | -1.0       | 11110001 | 4.0        | 11110011 | 4.0        |
| 11110101 | 4.0        | 11110111 | 4.0        | 11111001 | 5.0        | 11111011 | 4.0        |
| 11111101 | 4.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

### 1.17 Tables from Faces using OMP Regression model with 116 coefficients

Table 49: Faces, Length: 116, Numbers: Number 0.0: 139, Number 4.0: 24, Number 5.0: 23, Number -1.0: 14, Number 9.0: 10, Number 1.0: 8, Number 8.0: 8, Number 6.0: 8, Number 10.0: 5, Number 14.0: 3, Number 3.0: 3, Number 15.0: 2, Number -5.0: 2, Number 13.0: 2, Number 11.0: 1, Number 19.0: 1, Number -6.0: 1, Number -2.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001000 | 1.0        | 00001001 | 1.0        | 00001010 | 5.0        | 00001011 | 10.0       |
| 00001110 | 4.0        | 00001111 | 4.0        | 00010000 | 5.0        | 00010001 | 5.0        |
| 00010010 | 6.0        | 00010011 | 5.0        | 00010100 | 5.0        | 00010110 | 6.0        |
| 00011000 | 6.0        | 00011001 | 6.0        | 00011010 | 11.0       | 00011011 | 15.0       |
| 00011100 | 5.0        | 00011110 | 10.0       | 00011111 | 4.0        | 00100011 | -1.0       |
| 00100111 | -1.0       | 00101000 | 1.0        | 00101001 | 1.0        | 00101010 | 10.0       |
| 00101011 | 14.0       | 00101110 | 9.0        | 00101111 | 8.0        | 00110000 | 5.0        |
| 00110001 | 5.0        | 00110010 | 5.0        | 00110011 | 4.0        | 00110100 | 5.0        |
| 00110110 | 5.0        | 00110111 | -1.0       | 00111000 | 6.0        | 00111001 | 6.0        |
| 00111010 | 15.0       | 00111011 | 19.0       | 00111100 | 5.0        | 00111110 | 14.0       |
| 00111111 | 8.0        | 01001010 | 4.0        | 01001011 | 9.0        | 01001110 | 4.0        |
| 01001111 | 4.0        | 01010010 | 1.0        | 01010110 | 1.0        | 01011010 | 5.0        |
| 01011011 | 9.0        | 01011110 | 5.0        | 01011111 | 4.0        | 01100011 | -1.0       |
| 01100111 | -1.0       | 01101010 | 9.0        | 01101011 | 13.0       | 01101110 | 9.0        |
| 01101111 | 8.0        | 01110011 | -1.0       | 01110111 | -1.0       | 01111010 | 9.0        |
| 01111011 | 13.0       | 01111110 | 9.0        | 01111111 | 8.0        | 10001010 | 4.0        |
| 10001011 | 9.0        | 10001110 | 4.0        | 10001111 | 4.0        | 10010000 | 5.0        |
| 10010001 | 5.0        | 10010010 | 6.0        | 10010011 | 5.0        | 10010100 | 5.0        |
| 10010110 | 6.0        | 10011000 | 5.0        | 10011001 | 5.0        | 10011010 | 10.0       |
| 10011011 | 14.0       | 10011100 | 5.0        | 10011110 | 10.0       | 10011111 | 4.0        |
| 10100011 | -1.0       | 10100111 | -1.0       | 10101010 | 4.0        | 10101011 | 8.0        |
| 10101110 | 4.0        | 10101111 | 3.0        | 10110011 | -1.0       | 10110101 | -5.0       |
| 10110111 | -6.0       | 10111010 | 4.0        | 10111011 | 8.0        | 10111101 | -5.0       |
| 10111110 | 4.0        | 10111111 | -2.0       | 11001010 | 4.0        | 11001011 | 9.0        |
| 11001110 | 4.0        | 11001111 | 4.0        | 11010010 | 1.0        | 11010110 | 1.0        |
| 11011010 | 5.0        | 11011011 | 9.0        | 11011110 | 5.0        | 11011111 | 4.0        |
| 11100011 | -1.0       | 11100111 | -1.0       | 11101010 | 4.0        | 11101011 | 8.0        |
| 11101110 | 4.0        | 11101111 | 3.0        | 11110011 | -1.0       | 11110111 | -1.0       |
| 11111010 | 4.0        | 11111011 | 8.0        | 11111110 | 4.0        | 11111111 | 3.0        |
| Others   | 0.0        |          |            |          |            |          |            |



Table 50: Faces, Length: 116, Numbers: Number 0.0: 139, Number 5.0: 22, Number 9.0: 19, Number 4.0: 15, Number -1.0: 14, Number 3.0: 13, Number 6.0: 8, Number 1.0: 8, Number 14.0: 4, Number 10.0: 4, Number 15.0: 3, Number -5.0: 2, Number -6.0: 1, Number 11.0: 1, Number 20.0: 1, Number -2.0: 1

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00000010 | 5.0        | 00000110 | 5.0        | 00001010 | 5.0        | 00010010 | 5.0        |
| 00010110 | 5.0        | 00011010 | 5.0        | 00100010 | 5.0        | 00100110 | 5.0        |
| 00101010 | 5.0        | 00111110 | -5.0       | 01000000 | 1.0        | 01000010 | 6.0        |
| 01000110 | 5.0        | 01001000 | 1.0        | 01001010 | 6.0        | 01010010 | 5.0        |
| 01010100 | -1.0       | 01010101 | -1.0       | 01010110 | 4.0        | 01010111 | -1.0       |
| 01011010 | 5.0        | 01011100 | -1.0       | 01011101 | -1.0       | 01011110 | -1.0       |
| 01011111 | -1.0       | 01100000 | 1.0        | 01100010 | 6.0        | 01100110 | 5.0        |
| 01101000 | 1.0        | 01101010 | 6.0        | 01110100 | -1.0       | 01110101 | -1.0       |
| 01110110 | -1.0       | 01110111 | -1.0       | 01111100 | -1.0       | 01111101 | -1.0       |
| 01111110 | -6.0       | 01111111 | -1.0       | 10000010 | 6.0        | 10000011 | 1.0        |
| 10000110 | 6.0        | 10000111 | 1.0        | 10001010 | 5.0        | 10010010 | 6.0        |
| 10010011 | 1.0        | 10010110 | 6.0        | 10010111 | 1.0        | 10011010 | 5.0        |
| 10100010 | 5.0        | 10100110 | 5.0        | 10101010 | 5.0        | 10111110 | -5.0       |
| 11000000 | 5.0        | 11000001 | 4.0        | 11000010 | 11.0       | 11000011 | 5.0        |
| 11000100 | 4.0        | 11000101 | 4.0        | 11000110 | 10.0       | 11000111 | 5.0        |
| 11001000 | 10.0       | 11001001 | 9.0        | 11001010 | 15.0       | 11001011 | 9.0        |
| 11001100 | 4.0        | 11001101 | 4.0        | 11001110 | 4.0        | 11001111 | 4.0        |
| 11010000 | 4.0        | 11010001 | 4.0        | 11010010 | 10.0       | 11010011 | 5.0        |
| 11010100 | 3.0        | 11010101 | 3.0        | 11010110 | 9.0        | 11010111 | 4.0        |
| 11011000 | 9.0        | 11011001 | 9.0        | 11011010 | 14.0       | 11011011 | 9.0        |
| 11011100 | 3.0        | 11011101 | 3.0        | 11011110 | 3.0        | 11011111 | 3.0        |
| 11100000 | 10.0       | 11100001 | 9.0        | 11100010 | 15.0       | 11100011 | 9.0        |
| 11100100 | 9.0        | 11100101 | 9.0        | 11100110 | 14.0       | 11100111 | 9.0        |
| 11101000 | 15.0       | 11101001 | 14.0       | 11101010 | 20.0       | 11101011 | 14.0       |
| 11101100 | 9.0        | 11101101 | 9.0        | 11101110 | 9.0        | 11101111 | 9.0        |
| 11110000 | 4.0        | 11110001 | 4.0        | 11110010 | 4.0        | 11110011 | 4.0        |
| 11110100 | 3.0        | 11110101 | 3.0        | 11110110 | 3.0        | 11110111 | 3.0        |
| 11111000 | 9.0        | 11111001 | 9.0        | 11111010 | 9.0        | 11111011 | 9.0        |
| 11111100 | 3.0        | 11111101 | 3.0        | 11111110 | -2.0       | 11111111 | 3.0        |
| Others   | 0.0        |          |            |          |            |          |            |

### 1.18 Tables from Tetravoxels using OMP Regression model with 38 coefficients

Table 51: Tetravoxels, Length: 38, Numbers: Number 0.0: 217, Number 1.0: 30, Number 2.0: 6, Number 3.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001111 | 1.0        | 00011111 | 1.0        | 00101111 | 1.0        | 00111111 | 1.0        |
| 01001111 | 1.0        | 01011111 | 1.0        | 01101111 | 1.0        | 01111111 | 1.0        |
| 10001111 | 1.0        | 10011111 | 1.0        | 10101010 | 1.0        | 10101011 | 1.0        |
| 10101110 | 1.0        | 10101111 | 2.0        | 10111010 | 1.0        | 10111011 | 1.0        |
| 10111110 | 1.0        | 10111111 | 2.0        | 11001100 | 1.0        | 11001101 | 1.0        |
| 11001110 | 1.0        | 11001111 | 2.0        | 11011100 | 1.0        | 11011101 | 1.0        |
| 11011110 | 1.0        | 11011111 | 2.0        | 11101010 | 1.0        | 11101011 | 1.0        |
| 11101100 | 1.0        | 11101101 | 1.0        | 11101110 | 2.0        | 11101111 | 3.0        |
| 11111010 | 1.0        | 11111011 | 1.0        | 11111100 | 1.0        | 11111101 | 1.0        |
| 11111110 | 2.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

Table 52: Tetravoxels, Length: 38, Numbers: Number 0.0: 217, Number 1.0: 30, Number 2.0: 6, Number 3.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00110011 | 1.0        | 00110111 | 1.0        | 00111011 | 1.0        | 00111111 | 1.0        |
| 01010101 | 1.0        | 01010111 | 1.0        | 01011101 | 1.0        | 01011111 | 1.0        |
| 01110011 | 1.0        | 01110101 | 1.0        | 01110111 | 2.0        | 01111011 | 1.0        |
| 01111101 | 1.0        | 01111111 | 2.0        | 10110011 | 1.0        | 10110111 | 1.0        |
| 10111011 | 1.0        | 10111111 | 1.0        | 11010101 | 1.0        | 11010111 | 1.0        |
| 11011101 | 1.0        | 11011111 | 1.0        | 11110000 | 1.0        | 11110001 | 1.0        |
| 11110010 | 1.0        | 11110011 | 2.0        | 11110100 | 1.0        | 11110101 | 2.0        |
| 11110110 | 1.0        | 11110111 | 3.0        | 11111000 | 1.0        | 11111001 | 1.0        |
| 11111010 | 1.0        | 11111011 | 2.0        | 11111100 | 1.0        | 11111101 | 2.0        |
| 11111110 | 1.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

Table 53: Tetravoxels, Length: 38, Numbers: Number 0.0: 217, Number 1.0: 30, Number 2.0: 6, Number 3.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001111 | 1.0        | 00011111 | 1.0        | 00101111 | 1.0        | 00111111 | 1.0        |
| 01001111 | 1.0        | 01010101 | 1.0        | 01010111 | 1.0        | 01011101 | 1.0        |
| 01011111 | 2.0        | 01101111 | 1.0        | 01110101 | 1.0        | 01110111 | 1.0        |
| 01111101 | 1.0        | 01111111 | 2.0        | 10001111 | 1.0        | 10011111 | 1.0        |
| 10101111 | 1.0        | 10111111 | 1.0        | 11001100 | 1.0        | 11001101 | 1.0        |
| 11001110 | 1.0        | 11001111 | 2.0        | 11010101 | 1.0        | 11010111 | 1.0        |
| 11011100 | 1.0        | 11011101 | 2.0        | 11011110 | 1.0        | 11011111 | 3.0        |
| 11101100 | 1.0        | 11101101 | 1.0        | 11101110 | 1.0        | 11101111 | 2.0        |
| 11110101 | 1.0        | 11110111 | 1.0        | 11111100 | 1.0        | 11111101 | 2.0        |
| 11111110 | 1.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

Table 54: Tetravoxels, Length: 38, Numbers: Number 0.0: 217, Number 1.0: 30, Number 2.0: 6, Number 3.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 01010101 | 1.0        | 01010111 | 1.0        | 01011101 | 1.0        | 01011111 | 1.0        |
| 01110101 | 1.0        | 01110111 | 1.0        | 01111101 | 1.0        | 01111111 | 1.0        |
| 11001100 | 1.0        | 11001101 | 1.0        | 11001110 | 1.0        | 11001111 | 1.0        |
| 11010101 | 1.0        | 11010111 | 1.0        | 11011100 | 1.0        | 11011101 | 2.0        |
| 11011110 | 1.0        | 11011111 | 2.0        | 11101100 | 1.0        | 11101101 | 1.0        |
| 11101110 | 1.0        | 11101111 | 1.0        | 11110000 | 1.0        | 11110001 | 1.0        |
| 11110010 | 1.0        | 11110011 | 1.0        | 11110100 | 1.0        | 11110101 | 2.0        |
| 11110110 | 1.0        | 11110111 | 2.0        | 11111000 | 1.0        | 11111001 | 1.0        |
| 11111010 | 1.0        | 11111011 | 1.0        | 11111100 | 2.0        | 11111101 | 3.0        |
| 11111110 | 2.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

Table 55: Tetravoxels, Length: 38, Numbers: Number 0.0: 217, Number 1.0: 30, Number 2.0: 6, Number 3.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00110011 | 1.0        | 00110111 | 1.0        | 00111011 | 1.0        | 00111111 | 1.0        |
| 01110011 | 1.0        | 01110111 | 1.0        | 01111011 | 1.0        | 01111111 | 1.0        |
| 10101010 | 1.0        | 10101011 | 1.0        | 10101110 | 1.0        | 10101111 | 1.0        |
| 10110011 | 1.0        | 10110111 | 1.0        | 10111010 | 1.0        | 10111011 | 2.0        |
| 10111110 | 1.0        | 10111111 | 2.0        | 11101010 | 1.0        | 11101011 | 1.0        |
| 11101110 | 1.0        | 11101111 | 1.0        | 11110000 | 1.0        | 11110001 | 1.0        |
| 11110010 | 1.0        | 11110011 | 2.0        | 11110100 | 1.0        | 11110101 | 1.0        |
| 11110110 | 1.0        | 11110111 | 2.0        | 11111000 | 1.0        | 11111001 | 1.0        |
| 11111010 | 2.0        | 11111011 | 3.0        | 11111100 | 1.0        | 11111101 | 1.0        |
| 11111110 | 2.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

Table 56: Tetravoxels, Length: 38, Numbers: Number 0.0: 217, Number 1.0: 30, Number 2.0: 6, Number 3.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 10101010 | 1.0        | 10101011 | 1.0        | 10101110 | 1.0        | 10101111 | 1.0        |
| 10111010 | 1.0        | 10111011 | 1.0        | 10111110 | 1.0        | 10111111 | 1.0        |
| 11001100 | 1.0        | 11001101 | 1.0        | 11001110 | 1.0        | 11001111 | 1.0        |
| 11011100 | 1.0        | 11011101 | 1.0        | 11011110 | 1.0        | 11011111 | 1.0        |
| 11101010 | 1.0        | 11101011 | 1.0        | 11101100 | 1.0        | 11101101 | 1.0        |
| 11101110 | 2.0        | 11101111 | 2.0        | 11110000 | 1.0        | 11110001 | 1.0        |
| 11110010 | 1.0        | 11110011 | 1.0        | 11110100 | 1.0        | 11110101 | 1.0        |
| 11110110 | 1.0        | 11110111 | 1.0        | 11111000 | 1.0        | 11111001 | 1.0        |
| 11111010 | 2.0        | 11111011 | 2.0        | 11111100 | 2.0        | 11111101 | 2.0        |
| 11111110 | 3.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

Table 57: Tetravoxels, Length: 38, Numbers: Number 0.0: 217, Number 1.0: 30, Number 2.0: 6, Number 3.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001111 | 1.0        | 00011111 | 1.0        | 00101111 | 1.0        | 00110011 | 1.0        |
| 00110111 | 1.0        | 00111011 | 1.0        | 00111111 | 2.0        | 01001111 | 1.0        |
| 01011111 | 1.0        | 01101111 | 1.0        | 01110011 | 1.0        | 01110111 | 1.0        |
| 01111011 | 1.0        | 01111111 | 2.0        | 10001111 | 1.0        | 10011111 | 1.0        |
| 10101010 | 1.0        | 10101011 | 1.0        | 10101110 | 1.0        | 10101111 | 2.0        |
| 10110011 | 1.0        | 10110111 | 1.0        | 10111010 | 1.0        | 10111011 | 2.0        |
| 10111110 | 1.0        | 10111111 | 3.0        | 11001111 | 1.0        | 11011111 | 1.0        |
| 11101010 | 1.0        | 11101011 | 1.0        | 11101110 | 1.0        | 11101111 | 2.0        |
| 11110011 | 1.0        | 11110111 | 1.0        | 11111010 | 1.0        | 11111011 | 2.0        |
| 11111110 | 1.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |

Table 58: Tetravoxels, Length: 38, Numbers: Number 0.0: 217, Number 1.0: 30, Number 2.0: 6, Number 3.0: 2

| Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ | Index    | $\Delta E$ |
|----------|------------|----------|------------|----------|------------|----------|------------|
| 00001111 | 1.0        | 00011111 | 1.0        | 00101111 | 1.0        | 00110011 | 1.0        |
| 00110111 | 1.0        | 00111011 | 1.0        | 00111111 | 2.0        | 01001111 | 1.0        |
| 01010101 | 1.0        | 01010111 | 1.0        | 01011101 | 1.0        | 01011111 | 2.0        |
| 01101111 | 1.0        | 01110011 | 1.0        | 01110101 | 1.0        | 01110111 | 2.0        |
| 01111011 | 1.0        | 01111101 | 1.0        | 01111111 | 3.0        | 10001111 | 1.0        |
| 10011111 | 1.0        | 10101111 | 1.0        | 10110011 | 1.0        | 10110111 | 1.0        |
| 10111011 | 1.0        | 10111111 | 2.0        | 11001111 | 1.0        | 11010101 | 1.0        |
| 11010111 | 1.0        | 11011101 | 1.0        | 11011111 | 2.0        | 11101111 | 1.0        |
| 11110011 | 1.0        | 11110101 | 1.0        | 11110111 | 2.0        | 11111011 | 1.0        |
| 11111101 | 1.0        | 11111111 | 3.0        | Others   | 0.0        |          |            |